

where $0 \neq a \in F$. Then $I_1 = Fx_1 + Fx_2 + Fy$ and $I_2 = Fx_1 + Fx_2 + Fz$ are different maximal nilpotent ideals in A . By choosing $a = 1$ or $a = -1$ we obtain a commutative or anticommutative algebra A .

3. In general, in a nonassociative algebra, a power of an element is not uniquely determined. In the previous example, for the element $w = x_1 + x_2 + y + z$ we have

$$w^2w^2 = 0 \quad \text{but} \quad w(ww^2) = (1+a)(x_1 + x_2).$$

4. Let $A_1, \dots, A_n$ be simple algebras over a field F with bases

$$\{v_i^1 \mid i \in I_1\}, \dots, \{v_i^n \mid i \in I_n\}.$$

Consider the algebra $A = Fe \oplus A_1 \oplus \dots \oplus A_n$ (vector space direct sum) with multiplication defined by the following conditions:

- (a) The A_i are subalgebras of A .
- (b) $A_i A_j = \{0\}$ for $i \neq j$.
- (c) $ev_i^j = v_i^j e = e$ for all i, j .
- (d) $e^2 = e$.

Then $I = Fe$ is the unique minimal ideal in A , and $I^2 = I$. In particular, $\text{Solv } A = \{0\}$, but A is not semisimple (compare with the Lie algebra case).

5. **Suttles' example.** (Notices AMS 19 (1972) A-566) Let A be a commutative algebra over a field F of characteristic $\neq 2$, with basis x_i ($1 \leq i \leq 5$) and the following multiplication table (all other products are zero):

$$x_1x_2 = x_2x_4 = -x_1x_5 = x_3, \quad x_1x_3 = x_4, \quad x_2x_3 = x_5.$$

Then A is a solvable power associative nil algebra that is not nilpotent. Moreover, $\text{Nil } A = \text{Solv } A = A$, and $\text{Nilp } A$ does not exist (if F is infinite then A has infinitely many maximal nilpotent ideals).

69.3 Composition Algebras

Definitions:

A **composition algebra** is an algebra A with unit 1 over a field F of characteristic $\neq 2$ together with a **norm** $n(x)$ (a nondegenerate quadratic form on the vector space A) that **admits composition** in the sense that

$$n(xy) = n(x)n(y) \quad \text{for all } x \in A.$$

A **quadratic algebra** A over a field F is a unital algebra in which every $x \in A$ satisfies the condition $x^2 \in \text{Span}(x, 1)$. In other words, every subalgebra of A generated by a single element has dimension ≤ 2 .

A composition algebra A is **split** if it contains zero-divisors, that is, if $xy = 0$ for some nonzero $x, y \in A$.

Facts: ([Sch66], [Jac68], [ZSS82], [Bae02])

1. Every composition algebra A is alternative and quadratic. Moreover, every element $x \in A$ satisfies the equation

$$x^2 - t(x)x + n(x) = 0,$$

where $t(x)$ is a linear form on A (the **trace**) and $n(x)$ is the original quadratic form on A (the **norm**).

2. For a composition algebra A the following conditions are equivalent:
 - (a) A is split;
 - (b) $n(x) = 0$ for some nonzero $x \in A$;
 - (c) A contains an idempotent $e \neq 1$.
3. Let A be a unital algebra over a field F with an involution $x \mapsto \bar{x}$ satisfying Equation 69.1 from Example 2 of Section 69.1. The Cayley–Dickson doubling process gives the algebra (A, a) defined by Equations 69.2 to 69.4. It is clear that A is isomorphically embedded into (A, a) and that $\dim(A, a) = 2 \dim A$. For $v = (0, 1)$, we have $v^2 = a$ and $(A, a) = A \oplus Av$. For any $y = y_1 + y_2v \in (A, a)$, we have $\bar{y} = \overline{y_1} - y_2v$.
4. In a composition algebra A , the mapping $x \mapsto \bar{x} = t(x) - x$ is an involution of A fixing the elements of the field $F = F \cdot 1$. Conversely, if A is an alternative algebra with unit 1 and involution $x \mapsto \bar{x}$ satisfying Equation 69.1 from Example 2 of Section 69.1, then $x\bar{x} \in F$ and the quadratic form $n(x) = x\bar{x}$ satisfies $n(xy) = n(x)n(y)$.
5. The mapping $y \mapsto \bar{y}$ is an involution of (A, a) extending the involution $x \mapsto \bar{x}$ of A . Moreover, $y + \bar{y}$ and $y\bar{y}$ are in F for every $y \in (A, a)$. If the quadratic form $n(x) = x\bar{x}$ is nondegenerate on A , then the quadratic form $n(y) = y\bar{y}$ is nondegenerate on (A, a) , and the form $n(y)$ admits composition on (A, a) if and only if A is associative.
6. Every composition algebra over a field F of characteristic $\neq 2$ is isomorphic to F or to one of the algebras of types 2a–2c obtained from F by the Cayley–Dickson process as in Example 2 of Section 69.1.
7. Every split composition algebra over a field F is isomorphic to one of the algebras $F \oplus F$, $M_2(F)$, $\text{Zorn}(F)$ described in Examples 2 to 4 below.
8. Every finite dimensional composition algebra without zero divisors is a division algebra, and so every composition algebra is either split or a division algebra.
9. Every composition algebra of dimension > 1 over an algebraically closed field is split, and so every composition algebra over an algebraically closed field F is isomorphic to one of the algebras F , $F \oplus F$, $M_2(F)$, $\text{Zorn}(F)$.

Examples:

1. The fields of real numbers $\mathbb{R}$ and complex numbers $\mathbb{C}$, the quaternions $\mathbb{H}$, and the octonions $\mathbb{O}$, are real composition algebras with the Euclidean norm $n(x) = x\bar{x}$. The first three are associative; the algebra $\mathbb{O}$ provides us with the first and most important example of a nonassociative alternative algebra.
2. Let F be a field and let $A = F \oplus F$ be the direct sum of two copies of the field with the exchange involution $\overline{(a, b)} = (b, a)$, the trace $t((a, b)) = a + b$, and the norm $n((a, b)) = ab$. Then A is a two-dimensional split composition algebra.
3. Let $A = M_2(F)$ be the algebra of 2×2 matrices over F with the symplectic involution

$$x = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto \bar{x} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

the matrix trace $t(x) = a + d$, and the determinant norm $n(x) = ad - bc$. Then A is a 4-dimensional split composition algebra.

4. An eight-dimensional split composition algebra is the **Zorn vector-matrix algebra** (or the **Cayley–Dickson matrix algebra**), obtained by taking $A = \text{Zorn}(F)$, which consists of all 2×2 block matrices with scalars on the diagonal and 3×1 column vectors off the diagonal

$$\text{Zorn}(F) = \left\{ x = \begin{bmatrix} a & \mathbf{u} \\ \mathbf{v} & b \end{bmatrix} \mid a, b \in F, \mathbf{u}, \mathbf{v} \in F^3 \right\},$$

with norm, involution, and product

$$n(x) = ab - (\mathbf{u}, \mathbf{v}), \quad \bar{x} = \begin{bmatrix} b & -\mathbf{u} \\ -\mathbf{v} & a \end{bmatrix},$$

$$x_1 x_2 = \begin{bmatrix} a_1 a_2 + (\mathbf{u}_1, \mathbf{v}_2) & a_1 \mathbf{u}_2 + \mathbf{u}_1 b_2 - \mathbf{v}_1 \times \mathbf{v}_2 \\ \mathbf{v}_1 a_2 + b_1 \mathbf{v}_2 + \mathbf{u}_1 \times \mathbf{u}_2 & b_1 b_2 + (\mathbf{v}_1, \mathbf{u}_2) \end{bmatrix};$$

the scalar and vector products are defined for $\mathbf{u} = [u_1, u_2, u_3]^T$ and $\mathbf{v} = [v_1, v_2, v_3]^T$ by

$$(\mathbf{u}, \mathbf{v}) = u_1 v_1 + u_2 v_2 + u_3 v_3, \quad \mathbf{u} \times \mathbf{v} = [u_2 v_3 - u_3 v_2, u_3 v_1 - u_1 v_3, u_1 v_2 - u_2 v_1].$$

Applications:

1. If we write the equation $n(x)n(y) = n(xy)$ in terms of the coefficients of the algebra elements x, y with respect to an orthogonal basis for each of the composition algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ given in Example 3 of Section 69.1, then we obtain an identity expressing the multiplicativity of a quadratic form:

$$(x_1^2 + \cdots + x_k^2)(y_1^2 + \cdots + y_k^2) = z_1^2 + \cdots + z_k^2.$$

Here the z_i are bilinear functions in the x_i and y_i : To be precise, z_i is the coefficient of the i th basis vector in the product of the elements $x = (x_1, \dots, x_k)$ and $y = (y_1, \dots, y_k)$. By Hurwitz' theorem, such a k -square identity exists only for $k = 1, 2, 4, 8$.

69.4 Alternative Algebras

Definitions:

A **left alternative algebra** is one satisfying the identity $(x, x, y) = 0$.

A **right alternative algebra** is one satisfying the identity $(y, x, x) = 0$.

A **flexible algebra** is one satisfying the identity $(x, y, x) = 0$.

An **alternative algebra** is one satisfying all three identities (any two imply the third).

The **Moufang identities** play an important role in the theory of alternative algebras:

$(xy \cdot z)y = x(yzy)$	right Moufang identity
$(yzy)x = y(z \cdot yx)$	left Moufang identity
$(xy)(zx) = x(yz)x$	central Moufang identity.

(The terms yzy and $x(yz)x$ are well-defined by the flexible identity.)

An **alternative bimodule** over an alternative algebra A is an A -bimodule M for which the split null extension $E(A, M)$ is alternative.

Let A be an alternative algebra, let M be an alternative A -bimodule, and let (λ, ρ) be the associated birepresentation of A . The algebra A **acts nilpotently** on M if the subalgebra of $\text{End } M$, which is generated by the elements $\lambda(x), \rho(x)$ for all $x \in A$, is nilpotent. If $y \in A$, then y **acts nilpotently** on M if the elements $\lambda(y), \rho(y)$ generate a nilpotent subalgebra of $\text{End } M$.

A finite dimensional alternative algebra A over a field F is **separable** if the algebra $A_K = K \otimes_F A$ is nil-semisimple for any extension K of the field F .

Facts: ([Sch66], [ZSS82]) Additional facts about alternative algebras are given in Section 69.1 and facts about right alternative algebras are given in Section 69.6:

1. Every commutative or anticommutative algebra is flexible.
2. Substituting $x + z$ for x in the left alternative identity, and using distributivity, we obtain

$$(x, z, y) + (z, x, y) = 0.$$

This is the **linearization on x** of the left alternative identity. Linearizing the right alternative identity in the same way, we get

$$(y, x, z) + (y, z, x) = 0.$$

From the last two identities, it follows that in any alternative algebra A the associator (x, y, z) is a skew-symmetric (alternating) function of the arguments x, y, z .

3. Every alternative algebra is power associative (Corollary of Artin's Theorem, Fact 7, Section 69.1). In particular, the nil radical $\text{Nil } A$ exists in the class of alternative algebras.
4. Every alternative algebra satisfies the three Moufang identities and the identities

$$(x, y, yz) = (x, y, z)y, \quad (x, y, zy) = y(x, y, z).$$

5. A bimodule M over an alternative algebra A is alternative if and only if the following relations hold in the split null extension $E(A, M)$:

$$(x, m, x) = 0 \quad \text{and} \quad (x, m, y) = (m, y, x) = (y, x, m) \quad \text{for all } x, y \in A \quad \text{and} \quad m \in M.$$

6. It follows from the definition of alternative bimodule and the Moufang identities that

$$[\rho(x), \lambda(x)] = 0, \quad \rho(x^k) = (\rho(x))^k \quad \text{for } k \geq 1, \quad \lambda(x^k) = (\lambda(x))^k \quad \text{for } k \geq 1.$$

This implies that any nilpotent element of an alternative algebra acts nilpotently on any bimodule.

7. If every element of an alternative algebra A acts nilpotently on a finite dimensional alternative A -bimodule M , then A acts nilpotently on M .
8. A nilpotent algebra A acts nilpotently on the A -bimodule M if and only if the algebra $E(A, M)$ is nilpotent.
9. In a finite dimensional alternative algebra A , every nil subalgebra is nilpotent. In particular, the nil radical $\text{Nil } A$ is nilpotent.
10. The subclass Nilp of nilpotent algebras is radical in the class of all finite dimensional alternative algebras. For any finite dimensional alternative algebra A , we have

$$\text{Nil } A = \text{Solv } A = \text{Nilp } A.$$

11. Let A be a finite dimensional alternative algebra. The quotient algebra $A/\text{Nil } A$ is semisimple, that is, it decomposes into a direct sum of simple algebras. Every finite dimensional nil-semisimple alternative algebra is isomorphic to a direct sum of simple algebras, where every simple algebra is either a matrix algebra over a skew-field or a Cayley–Dickson algebra over its center.
12. Let A be a finite dimensional alternative algebra over a field F . If the quotient algebra $A/\text{Nil } A$ is separable over F , then there exists a subalgebra B of A such that B is isomorphic to $A/\text{Nil } A$ and $A = B \oplus \text{Nil } A$ (vector space direct sum).
13. Every alternative bimodule over a separable alternative algebra is completely reducible (as in the case of associative algebras).
14. Let A be a finite dimensional alternative algebra, let M be a faithful irreducible A -bimodule, and let (λ, ρ) be the associated birepresentation of A . Either M is an associative bimodule over A (which must then be associative), or one of the following holds:

- (a) The algebra A is an algebra of generalized quaternions, λ is a (right) associative irreducible representation of A , and $\rho(x) = \lambda(\bar{x})$ for every $x \in A$.
 - (b) The algebra $A = \mathbb{O}$ is a Cayley–Dickson algebra and M is isomorphic to $\text{Reg}(\mathbb{O})$.
15. Every simple alternative algebra (of any dimension) is either associative or is isomorphic to a Cayley–Dickson algebra over its center.

69.5 Jordan Algebras

In this section, we assume that the base field F has characteristic $\neq 2$.

Definitions:

A **Jordan algebra** is a commutative nonassociative algebra satisfying the **Jordan identity**

$$(x^2 y)x = x^2(yx).$$

The **linearization on x** of the Jordan identity is

$$2((xz)y)x + (x^2 y)z = 2(xz)(xy) + x^2(yz).$$

A Jordan algebra J is **special** if it is isomorphic to a subalgebra of the algebra A^+ for some associative algebra A ; otherwise, it is **exceptional**.

Facts: ([BK66], [Jac68], [ZSS82], [McC04]) Additional facts about Jordan algebras are given in Section 69.1, and facts about noncommutative Jordan algebras are given in Section 69.6:

1. (Zelmanov’s Simple Theorem) Every simple Jordan algebra (of any dimension) is isomorphic to one of the following: (a) an algebra of a bilinear form, (b) an algebra of Hermitian type, (c) an Albert algebra. For definitions see Examples 3, 4, and 5 below.
2. Let J be a Jordan algebra. Consider the regular birepresentation $x \mapsto L_x, x \mapsto R_x$ of the algebra J . Commutativity and the Jordan identity imply that for all $x, y \in J$ we have

$$L_x = R_x, \quad [R_x, R_{x^2}] = 0, \quad R_{x^2 y} - R_y R_{x^2} + 2R_x R_y R_x - 2R_x R_{yx} = 0.$$

Linearizing the last equation on x we see that for all $x, y, z \in J$ we have

$$R_{(xz)y} - R_y R_{xz} + R_x R_y R_z + R_z R_y R_x - R_x R_{yz} - R_z R_{yx} = 0.$$

3. For every $k \geq 1$, the operator R_{x^k} belongs to the subalgebra $A \subseteq \text{End } J$ generated by R_x and R_{x^2} . Since A is commutative, we have $[R_{x^k}, R_{x^\ell}] = 0$ for all $k, \ell \geq 1$, which can be written as $(x^k, J, x^\ell) = \{0\}$.
4. It follows from the previous fact that every Jordan algebra is power associative and the radical $\text{Nil } J$ is defined.
5. Let J be a finite dimensional Jordan algebra. As for alternative algebras, we have

$$\text{Nil } J = \text{Solv } J = \text{Nilp } J,$$

that is, the radical $\text{Nil } J$ is nilpotent. The quotient algebra $J/\text{Nil } J$ is semisimple, that is, isomorphic to a direct sum of simple algebras. If the quotient algebra $J/\text{Nil } J$ is separable over F , then there exists a subalgebra B of J such that B is isomorphic to $J/\text{Nil } J$ and $J = B \oplus \text{Nil } J$ (vector space direct sum).

6. If a Jordan algebra J contains an idempotent e , the operator R_e satisfies the equation $R_e(2R_e - 1)(R_e - 1) = 0$, and the algebra J has the following analogue of the Pierce decomposition from the theory of associative algebras:

$$J = J_1 \oplus J_{1/2} \oplus J_0, \quad \text{where} \quad J_i = J_i(e) = \{x \in J \mid xe = ix\}.$$

For $i, j = 0, 1$ ($i \neq j$), we have the inclusions

$$J_i^2 \subseteq J_i, \quad J_i J_{1/2} \subseteq J_{1/2}, \quad J_i J_j = \{0\}, \quad J_{1/2}^2 \subseteq J_1 + J_2.$$

More generally, if J has unit $1 = \sum_{i=1}^n e_i$ where e_i are orthogonal idempotents, then

$$J = \bigoplus_{i \leq j} J_{ij}, \quad \text{where} \quad J_{ii} = J_1(e_i), \quad J_{ij} = J_{1/2}(e_i) \cap J_{1/2}(e_j) \quad \text{for } i \neq j,$$

and the components J_{ij} are multiplied according to the rules

$$\begin{aligned} J_{ii}^2 &\subseteq J_{ii}, & J_{ij} J_{ii} &\subseteq J_{ij}, & J_{ij}^2 &\subseteq J_{ii} + J_{jj}, & J_{ii} J_{jj} &= \{0\} & \text{for distinct } i, j; \\ J_{ij} J_{jk} &\subseteq J_{ik}, & J_{ij} J_{kk} &= \{0\}, & J_{ij} J_{k\ell} &= \{0\} & \text{for distinct } i, j, k, \ell. \end{aligned}$$

7. Every Jordan algebra that contains >3 strongly connected orthogonal idempotents is special. (Orthogonal idempotents e_1, e_2 are **strongly connected** if there exists an element $u_{12} \in J_{12}$ for which $u_{12}^2 = e_1 + e_2$.)
8. (Coordinatization Theorem) Let J be a Jordan algebra with unit $1 = \sum_{i=1}^n e_i$ ($n \geq 3$), where the e_i are mutually strongly connected orthogonal idempotents. Then J is isomorphic to the Jordan algebra $H_n(D)$ of Hermitian $n \times n$ matrices over an alternative algebra D (which is associative for $n > 3$) with involution $*$ such that $H(D, *) \subseteq N(D)$, where $N(D)$ is the associative center of D .
9. Every Jordan bimodule over a separable Jordan algebra is completely reducible, and the structure of irreducible bimodules is known.

Examples:

- The algebra A^+ .** If A is an associative algebra, then the algebra A^+ is a Jordan algebra. Every subspace J of A closed with respect to the operation $x \cdot y = \frac{1}{2}(xy + yx)$ is a subalgebra of the algebra A^+ and every special Jordan algebra J is (up to isomorphism) of this type. The subalgebra of A generated by J is called the **associative enveloping algebra** of J . Properties of the algebras A and A^+ are closely related: A is simple (respectively nilpotent) if and only if A^+ is simple (respectively nilpotent).
- The algebra A^+ may be a Jordan algebra for nonassociative A ; for instance, if A is a right alternative (in particular, alternative) algebra, then A^+ is a special Jordan algebra.
- The algebra of a bilinear form.** Let X be a vector space of dimension > 1 over F , with a symmetric nondegenerate bilinear form $f(x, y)$. Consider the vector space direct sum $J(X, f) = F \oplus X$, and define on it a multiplication by assuming that the unit element $1 \in F$ is the unit element of $J(X, f)$ and by setting $xy = f(x, y)1$ for any $x, y \in X$. Then $J(X, f)$ is a simple special Jordan algebra; its associative enveloping algebra is the Clifford algebra $C(X, f)$ of the bilinear form f . When $F = \mathbb{R}$ and $f(x, y)$ is the ordinary dot product on X , the algebra $J(X, f)$ is called a **spin-factor**.
- Algebras of Hermitian type.** Let A be an associative algebra with involution $*$. The subspace $H(A, *) = \{x \in A \mid x^* = x\}$ of $*$ -symmetric elements is closed with respect to the Jordan multiplication $x \cdot y$ and, therefore, is a special Jordan algebra. For example, let D be an associative composition algebra with involution $x \mapsto \bar{x}$ and let $D^{n \times n}$ be the algebra of $n \times n$ matrices over D . Then the mapping $S: (x_{ij}) \mapsto (\overline{x_{ji}})$ is an involution of $D^{n \times n}$ and the set of D -Hermitian matrices $H_n(D) = H(D^{n \times n}, S)$ is a special Jordan algebra. If A is $*$ -simple (if it contains no proper ideal I with $I^* \subseteq I$), then $H(A, *)$ is simple. In particular, all the algebras $H_n(D)$ are simple. Every

algebra A^+ is isomorphic to the algebra $H(B, *)$ where $B = A \oplus A^{\text{opp}}$ (algebra direct sum) and $(x_1, x_2)^* = (x_2, x_1)$.

5. **Albert algebras.** If $D = \mathbb{O}$ is a Cayley–Dickson algebra, then the algebra $H_n(\mathbb{O})$ of Hermitian matrices over D is a Jordan algebra only for $n \leq 3$. For $n = 1, 2$ the algebras are isomorphic to algebras of bilinear forms and are, therefore, special. The algebra $H_3(\mathbb{O})$ is exceptional (not special). An algebra J is called an Albert algebra if $K \otimes_F J \cong H_3(\mathbb{O})$ for some extension K of the field F . Every Albert algebra is simple, exceptional, and has dimension 27 over its center.

69.6 Power Associative Algebras, Noncommutative Jordan Algebras, and Right Alternative Algebras

A natural generalization of Jordan algebras is the class of algebras that satisfy the Jordan identity but which are not necessarily commutative. If the algebra has a unit element, then the Jordan identity easily implies the flexible identity. The right alternative algebras have been the most studied among the power associative algebras that do not satisfy the flexible identity.

As in the previous section, we assume that F is a field of characteristic $\neq 2$.

Definitions:

A **noncommutative Jordan algebra** is an algebra satisfying the flexible and Jordan identities. In this definition the Jordan identity may be replaced by any of the identities

$$x^2(xy) = x(x^2y), \quad (yx)x^2 = (yx^2)x, \quad (xy)x^2 = (x^2y)x.$$

A subspace V of an algebra A is **right nilpotent** if $V^{(n)} = \{0\}$ for some $n \geq 1$, where $V^{(1)} = V$ and $V^{(n+1)} = V^{(n)}V$.

Facts: ([Sch66], [ZSS82], [KS95])

1. Let A be a finite dimensional power associative algebra with a bilinear symmetric form (x, y) satisfying the following conditions:
 - (a) $(xy, z) = (x, yz)$ for all $x, y, z \in A$.
 - (b) $(e, e) \neq 0$ for every idempotent $e \in A$.
 - (c) $(x, y) = 0$ if the product xy is nilpotent.

Then $\text{Nil } A = \text{Nil } A^+ = \{x \in A \mid (x, A) = \{0\}\}$, and if F has characteristic $\neq 2, 3, 5$, then the quotient algebra $A/\text{Nil } A$ is a noncommutative Jordan algebra.

2. Let A be a finite dimensional nil-semisimple flexible power associative algebra over an infinite field of characteristic $\neq 2, 3$. Then A has a unit element and is a direct sum of simple algebras, each of which is either a noncommutative Jordan algebra or (in the case of positive characteristic) an algebra of degree 2.
3. The structure of arbitrary finite dimensional nil-semisimple power associative algebras is still unclear. In particular, it is not known whether they are semisimple. It is known that in this case new simple algebras arise even in characteristic zero.
4. An algebra A is a noncommutative Jordan algebra if and only if it is flexible and the corresponding plus-algebra A^+ is a Jordan algebra.
5. Let A be a noncommutative Jordan algebra. For $x \in A$, the operators R_x, L_x, L_{x^2} generate a commutative subalgebra in the multiplication algebra $M(A)$, containing all the operators R_{x^k} and L_{x^k} for $k \geq 1$.

6. Every noncommutative Jordan algebra is power associative.
7. Let A be a finite dimensional nil-semisimple noncommutative Jordan algebra over F . Then A has a unit element and is a direct sum of simple algebras. If F has characteristic 0, then every simple summand is either a (commutative) Jordan algebra, a quasi-associative algebra (see Example 3 below), or a quadratic flexible algebra. In the case of positive characteristic, there are more examples of simple noncommutative Jordan algebras.
8. Unlike alternative and Jordan algebras, an analogue of the Wedderburn Principal Theorem on splitting of the nil radical does not hold in general for noncommutative Jordan algebras.
9. Every quasi-associative algebra (see Example 3 below) is a noncommutative Jordan algebra.
10. Every flexible quadratic algebra is a noncommutative Jordan algebra.
11. The right multiplication operators in every right alternative algebra A satisfy

$$R_{x^2} = R_x^2, \quad R_{x \cdot y} = R_x \cdot R_y,$$

where $x \cdot y = \frac{1}{2}(xy + yx)$ is the multiplication in the algebra A^+ . (Recall that, in this section, we assume that the characteristic of the base field is $\neq 2$.)

12. If A is a right alternative algebra (respectively, noncommutative Jordan algebra), then its unital hull $A^\#$ is also right alternative (respectively, noncommutative Jordan).
13. If A is a right alternative algebra, then the mapping $x \mapsto R_x$ is a homomorphism of the algebra A^+ into the special Jordan algebra $R(A)^+$. If A has a unit element, then this mapping is injective. For every right alternative algebra A , the algebra A^+ is embedded into the algebra $R(A^\#)^+$ and, hence, is a special Jordan algebra.
14. Every right alternative algebra A satisfies the identity

$$R_x^k R_x^\ell = R_{x^{k+\ell}} \quad \text{for any } x \in A \quad \text{and} \quad k, \ell \geq 1.$$

Therefore, A is power associative and the nil radical $\text{Nil } A$ is defined.

15. Let A be an arbitrary right alternative algebra. Then the quotient algebra $A/\text{Nil } A$ is alternative. In particular, every right alternative algebra without nilpotent elements is alternative.
16. Every simple right alternative algebra that is not a nil algebra is alternative (and, hence, either associative or a Cayley–Dickson algebra). The nonnil restriction is essential: There exists a nonalternative simple right alternative nil algebra.
17. A finite dimensional right alternative nil algebra is right nilpotent and, therefore, solvable, but such an algebra can be nonnilpotent. In particular, the subclass *Nilp* is not radical in the class of finite dimensional right alternative algebras.

Examples:

1. The Suttles algebra (Example 5 in Section 69.2) is a power associative algebra that is not a noncommutative Jordan algebra. For another example see Example 5 below.
2. The class of noncommutative Jordan algebras contains, apart from Jordan algebras, all alternative algebras (and, thus, all associative algebras) and all anticommutative algebras.
3. **Quasi-associative algebras.** Let A be an algebra over a field F and let $a \in F$, $a \neq \frac{1}{2}$. Define a new multiplication on A as follows:

$$x \cdot_a y = axy + (1 - a)yx,$$

and denote the resulting algebra by A^a . The passage from A to A^a is reversible: $A = (A^a)^b$ for $b = a/(2a - 1)$. Properties of A and A^a are closely related: The ideals (respectively subalgebras) of A are those of A^a ; the algebra A^a is nilpotent (respectively solvable, simple) if and only if the same holds for A . If A is associative, then A^a is a noncommutative Jordan algebra; furthermore, if the identity $[[x, y], z] = 0$ does not hold in A , then A^a is not associative. In particular, if A is

a simple noncommutative associative algebra, then A^a is an example of a simple nonassociative noncommutative Jordan algebra. The algebras of the form A^a for an associative algebra A are **split** quasi-associative algebras. More generally, an algebra A is quasi-associative if it has a scalar extension, which is a split quasi-associative algebra.

4. **Generalized Cayley–Dickson algebras.** For $a_1, \dots, a_n \in F - \{0\}$ let

$$A(a_1) = (F, a_1), \quad \dots, \quad A(a_1, \dots, a_n) = (A(a_1, \dots, a_{n-1}), a_n),$$

be the algebras obtained from F by successive application of the Cayley–Dickson process (Example 2 of section 69.1). Then $A(a_1, \dots, a_n)$ is a central simple quadratic noncommutative Jordan algebra of dimension 2^n .

5. Let V be a vector space of dimension $2n$ over a field F with a nondegenerate skew-symmetric bilinear form (x, y) . On the vector space direct sum $A = F \oplus V$ define a multiplication (as for the Jordan algebra of bilinear form) by letting the unit element 1 of F be the unit of A and by setting $xy = (x, y)1$ for any $x, y \in V$. Then A is a simple quadratic algebra (and, hence, it is power associative), but A is not flexible and, thus, is not a noncommutative Jordan algebra.

69.7 Malcev Algebras

Some of the theory of Malcev algebras generalizes the theory of Lie algebras. For information about Lie algebras, the reader is advised to consult Chapter 70.

Definitions:

A **Malcev algebra** is an anticommutative algebra satisfying the identity

$$J(x, y, xz) = J(x, y, z)x, \quad \text{where} \quad J(x, y, z) = (xy)z + (yz)x + (zx)y.$$

In a **left-normalized product** we omit the parenthesis, for example,

$$xyzx = ((xy)z)x, \quad yzxx = ((yz)x)x.$$

A **representation** of a Malcev algebra A is a linear mapping $\rho: A \rightarrow \text{End } V$ satisfying the following identity for all $x, y, z \in A$:

$$\rho(xy \cdot z) = \rho(x)\rho(y)\rho(z) - \rho(z)\rho(x)\rho(y) + \rho(y)\rho(zx) - \rho(yz)\rho(x).$$

We call V a **Malcev module** for A . The anticommutativity of A implies that the notion of a Malcev module is equivalent to that of Malcev bimodule; we set $xv = -vx$ for all $x \in A, v \in V$.

The **Killing form** $K(x, y)$ on a Malcev algebra A is defined (as for a Lie algebra) by

$$K(x, y) = \text{trace}(R_x R_y).$$

Facts: ([KS95], [She00], [PS04], [SZ06])

1. After expanding the Jacobians, the Malcev identity takes the form

$$xyzx + yzxx + zxx y = xy \cdot xz,$$

(using our convention on left-normalized products). If F has characteristic $\neq 2$, the Malcev identity is equivalent to the more symmetric identity

$$xyzt + yztx + ztxy + txyz = xz \cdot yt.$$

2. Any two elements in a Malcev algebra generate a Lie subalgebra.

3. The structure theory of finite dimensional Malcev algebras repeats the main features of the corresponding theory for Lie algebras. For any alternative algebra A , the minus algebra A^- is a Malcev algebra. Let $\mathbb{O} = \mathbb{O}(a, b, c)$ be a Cayley–Dickson algebra over a field F of characteristic $\neq 2$. Then $\mathbb{O} = F \oplus M$ (vector space direct sum), where $M = \{x \in \mathbb{O} \mid t(x) = 0\}$. The subspace M is a subalgebra (in fact, an ideal) of the Malcev algebra $\mathbb{O}^-$, and $M \cong \mathbb{O}^-/F$ (in fact, $\mathbb{O}^-$ is the Malcev algebra direct sum of the ideals F and M). The Malcev algebra $M = M(a, b, c)$ is central simple and has dimension 7 over F ; if F has characteristic $\neq 3$, then M is not a Lie algebra.
4. Every central simple Malcev algebra of characteristic $\neq 2$ is either a Lie algebra or an algebra $M(a, b, c)$. There are no non-Lie simple Malcev algebras in characteristic 3.
5. Two Malcev algebras $M(a, b, c)$, $M(a', b', c')$ are isomorphic if and only if the corresponding Cayley–Dickson algebras $\mathbb{O}(a, b, c)$, $\mathbb{O}(a', b', c')$ are isomorphic.
6. Let A be a finite dimensional Malcev algebra over a field F of characteristic 0 and let $\text{Solv } A$ be the solvable radical of A . The algebra A is semisimple (it decomposes into a direct sum of simple algebras) if and only if $\text{Solv } A = \{0\}$ (in fact, this is often used as the definition of “semisimple” for Malcev algebras, following the terminology for Lie algebras). If the quotient algebra $A/\text{Solv } A$ is separable, then A contains a subalgebra $B \cong A/\text{Solv } A$ and $A = B \oplus \text{Solv } A$ (vector space direct sum).
7. The Killing form $K(x, y)$ is symmetric and associative:

$$K(x, y) = K(y, x), \quad K(xy, z) = K(x, yz).$$

The algebra A is semisimple if and only if the form $K(x, y)$ is nondegenerate. For the solvable radical we have $\text{Solv } A = \{x \in A \mid K(x, A^2) = \{0\}\}$. In particular, A is solvable if and only if $K(A, A^2) = \{0\}$.

8. If all the operators $\rho(x)$ for $x \in A$ are nilpotent, then they generate a nilpotent subalgebra in $\text{End } V$ (A acts nilpotently on V). If the representation ρ is almost faithful (that is, $\ker \rho$ does not contain nonzero ideals of A), then A is nilpotent.
9. Every representation of a semisimple Malcev algebra A is completely reducible.
10. If A is a Malcev algebra and V is an A -bimodule, then V is a Malcev module for A if and only if the split null extension $E(A, V)$ is a Malcev algebra.
11. Let A be a Malcev algebra, and let V be a faithful irreducible A -module. Then the algebra A is simple, and either V is a Lie module over A (which must then be a Lie algebra) or one of the following holds:
 - (a) $A \cong M(a, b, c)$ and V is a regular A -module.
 - (b) A is isomorphic to the Lie algebra $sl(2, F)$ with $\dim V = 2$ and $\rho(x) = x^*$, where x^* is the matrix adjoint to $x \in A \subseteq M_2(F)$. (Here the matrix adjoint is defined by the equation $x x^* = x^* x = \det(x)I$, where I is the identity matrix.)
12. The **speciality problem** for Malcev algebras is still open: Is every Malcev algebra embeddable into the algebra A^- for some alternative algebra A ? This is the generalization of the Poincaré–Birkhoff–Witt theorem for Malcev algebras.

69.8 Akivis and Sabinin Algebras

The theory of Akivis and Sabinin algebras generalizes the theory of Lie algebras and their universal enveloping algebras. For information about Lie algebras, the reader is advised to consult Chapter 70.

Definitions:

An **Akivis algebra** is a vector space A over a field F , together with an anticommutative bilinear operation $A \times A \rightarrow A$ denoted $[x, y]$, and a trilinear operation $A \times A \times A \rightarrow A$ denoted (x, y, z) , satisfying the

Akivis identity for all $x, y, z \in A$:

$$[[x, y], z] + [[y, z], x] + [[z, x], y] = (x, y, z) + (y, z, x) + (z, x, y) - (x, z, y) - (y, x, z) - (z, y, x).$$

A **Sabinin algebra** is a vector space A over a field F , together with multilinear operations

$$\langle x_1, \dots, x_m; y, z \rangle \quad (m \geq 0),$$

satisfying these identities:

$$\langle x_1, \dots, x_m; y, y \rangle = 0, \quad (69.5)$$

$$\begin{aligned} & \langle x_1, \dots, x_r, u, v, x_{r+1}, \dots, x_m; y, z \rangle - \langle x_1, \dots, x_r, v, u, x_{r+1}, \dots, x_m; y, z \rangle \\ & + \sum_{k=0}^r \sum_s \langle x_{s(1)}, \dots, x_{s(k)}, \langle x_{s(k+1)}, \dots, x_{s(r)}; u, v \rangle, x_{r+1}, \dots, x_m; y, z \rangle = 0, \end{aligned} \quad (69.6)$$

$$K_{u,v,w} \left[\langle x_1, \dots, x_r, u; v, w \rangle + \sum_{k=0}^r \sum_s \langle x_{s(1)}, \dots, x_{s(k)}, \langle x_{s(k+1)}, \dots, x_{s(r)}; v, w \rangle, u \rangle \right] = 0, \quad (69.7)$$

where s is a $(k, r - k)$ -shuffle (a permutation of $1, \dots, r$ satisfying $s(1) < \dots < s(k)$ and $s(k+1) < \dots < s(r)$) and the operator $K_{u,v,w}$ denotes the sum over all cyclic permutations. (See Fact 9 below for an alternative formulation of this definition.)

An algebra $\mathcal{F}_{\mathcal{M}}[X]$ from a class $\mathcal{M}$, with a set of generators X , is called the **free algebra** in $\mathcal{M}$ with the set X of free generators, if any mapping of X into an arbitrary algebra $A \in \mathcal{M}$ extends uniquely to a homomorphism of $\mathcal{F}_{\mathcal{M}}[X]$ to A .

Let I be any subset of $\mathcal{F}_{\mathcal{M}}[X]$. The **T -ideal** in $\mathcal{F}_{\mathcal{M}}[X]$ determined by I , denoted by $T = T(I, X)$, is the smallest ideal of $\mathcal{F}_{\mathcal{M}}[X]$ containing all elements of the form $f(x_1, \dots, x_n)$ for all $f \in I$ and all $x_1, \dots, x_n \in \mathcal{F}_{\mathcal{M}}[X]$.

Facts: ([HS90], [She99], [SU02], [GH03], [Per05], [BHP05], [BDE05])

1. Free algebras may be constructed as follows. Let S be a set of generating elements and let Ω be a set of operation symbols. Let $r: \Omega \rightarrow \mathbb{N}$ (the nonnegative integers) be the arity function, that is, $\omega \in \Omega$ will represent an n -ary operation for $n = r(\omega)$. The set $W(S, \Omega)$ of nonassociative Ω -words on the set S is defined inductively as follows:

- (a) $S \subseteq W(S, \Omega)$.
- (b) If $\omega \in \Omega$ and $x_1, \dots, x_n \in W(S, \Omega)$ where $n = r(\omega)$, then $\omega(x_1, \dots, x_n) \in W(S, \Omega)$.

Let F be a field and let $F(S, \Omega)$ be the vector space over F with basis $W(S, \Omega)$. For each $\omega \in \Omega$ we define an n -ary operation with $n = r(\omega)$ on $F(S, \Omega)$, denoted by the same symbol ω , as follows: Given any basis elements $x_1, \dots, x_n \in W(S, \Omega)$, we set the value of ω on the arguments $x_1, \dots, x_n$ equal to the nonassociative word $\omega(x_1, \dots, x_n)$, and extend linearly to all of $F(S, \Omega)$. The algebra $F(S, \Omega)$ is the free Ω -algebra on the generating set S over the field F with the operations $\omega \in \Omega$.

2. The quotient algebra $F(S, \Omega)/T(I, S, \Omega)$ is the free $\mathcal{M}$ -algebra for the class $\mathcal{M} = \mathcal{M}(I)$ of Ω -algebras defined by the set of identities I .
3. Every subalgebra of a free Akivis algebra is again free.
4. Every Akivis algebra is isomorphic to a subalgebra of $\text{Akivis}(A)$ for some nonassociative algebra A . This generalizes the Poincaré–Birkhoff–Witt theorem for Lie algebras. The free nonassociative algebra is the universal enveloping algebra of the free Akivis algebra. (See Example 1 below.)
5. The free nonassociative algebra with generating set X has a natural structure of a (nonassociative) Hopf algebra, generalizing the Hopf algebra structure on the free associative algebra. The **Akivis elements** (the elements of the subalgebra generated by X using the commutator and associator) are properly contained in the **primitive elements** (the elements satisfying $\Delta(x) = x \otimes 1 + 1 \otimes x$ where Δ is the co-multiplication). The Akivis elements and the primitive elements have a natural

structure of an Akivis algebra. The primitive elements have the additional structure of a Sabinin algebra.

6. The Witt dimension formula for free Lie algebras (the primitive elements in the free associative algebra) has a generalization to the primitive elements in the free nonassociative algebra.
7. Sabinin algebras are a nonassociative generalization of Lie algebras in the following sense: The tangent space at the identity of any local analytic loop (without associativity assumptions) has a natural structure of a Sabinin algebra, and the classical correspondence between Lie groups and Lie algebras generalizes to this case.
8. Every Sabinin algebra arises as the subalgebra of primitive elements in some nonassociative Hopf algebra.
9. Another (equivalent) way to define Sabinin algebras, which exploits the Hopf algebra structure, is as follows. Let A be a vector space and let $T(A)$ be the tensor algebra of A . We write $\Delta: T(A) \rightarrow T(A) \otimes T(A)$ for the co-multiplication on $T(A)$: the algebra homomorphism that extends the diagonal mapping $\Delta: u \mapsto 1 \otimes u + u \otimes 1$ for $u \in A$. We will use the Sweedler notation and write $\Delta(x) = \sum x_{(1)} \otimes x_{(2)}$ for any $x \in T(A)$. Then A is a Sabinin algebra if it is equipped with a trilinear mapping

$$T(A) \otimes A \otimes A \rightarrow A, \quad x \otimes y \otimes z \mapsto \langle x; y, z \rangle, \quad \text{for } x \in T(A) \text{ and } y, z \in A,$$

satisfying the identities

$$\langle x; y, y \rangle = 0, \tag{69.8}$$

$$\begin{aligned} \langle x \otimes u \otimes v \otimes x'; y, z \rangle - \langle x \otimes v \otimes u \otimes x'; y, z \rangle + \sum \langle x_{(1)} \otimes \langle x_{(2)}; u, v \rangle \otimes x'; y, z \rangle \\ = 0, \end{aligned} \tag{69.9}$$

$$K_{u,v,w} \left[\langle x \otimes u; v, w \rangle + \sum \langle x_{(1)}; \langle x_{(2)}; v, w \rangle, u \rangle \right] = 0, \tag{69.10}$$

where $x, x' \in T(A)$ and $u, v, w, y, z \in A$. Identities (8 to 10) exploit the Sweedler notation to express identities (5 to 7) in a more compact form.

Examples:

1. Any nonassociative algebra A becomes an Akivis algebra $Akivis(A)$ if we define $[x, y]$ and (x, y, z) to be the commutator $xy - yx$ and the associator $(xy)z - x(yz)$. If A is an associative algebra, then the trilinear operation of $Akivis(A)$ is identically zero; in this case the Akivis identity reduces to the Jacobi identity, and so $Akivis(A)$ is a Lie algebra. If A is an alternative algebra, then the alternating property of the associator shows that the right side of the Akivis identity reduces to $6(x, y, z)$.
2. Every Lie algebra is an Akivis algebra with the identically zero trilinear operation. Every Malcev algebra (over a field of characteristic $\neq 2, 3$) is an Akivis algebra with the trilinear operation equal to $\frac{1}{6}J(x, y, z)$.
3. Every Akivis algebra A is a Sabinin algebra if we define

$$\langle a, b \rangle = -[a, b], \quad \langle x; a, b \rangle = (x, b, a) - (x, a, b), \quad \langle x_1, \dots, x_m; a, b \rangle = 0 \ (m > 1),$$

for all $a, b, x, x_i \in A$.

4. Let L be a Lie algebra with a subalgebra $H \subseteq L$ and a subspace $V \subseteq L$ for which $L = H \oplus V$. We write $P_V: L \rightarrow V$ for the projection onto V with respect to this decomposition of L . We define an operation

$$\langle -, -; - \rangle: T(V) \otimes V \otimes V \rightarrow V$$

by (using the Sweedler notation again)

$$\{x \otimes a \otimes b\} + \sum \{x_{(1)} \otimes \langle x_{(2)}; a, b \rangle\} = 0,$$

where for $x = x_1 \otimes \cdots \otimes x_n \in T(V)$ we write

$$\{x\} = P_V([x_1, [\dots, [x_{n-1}, x_n]] \cdots]).$$

Then the vector space V together with the operation $\langle -, -; - \rangle$ is a Sabinin algebra, and every Sabinin algebra can be obtained in this way.

69.9 Computational Methods

For homogeneous multilinear polynomial identities of degree n , the number of associative monomials is $n!$ and the number of association types is C_n (the Catalan number); hence, the number of nonassociative monomials grows superexponentially:

$$n! \cdot \frac{1}{n} \binom{2n-2}{n-1} = \frac{(2n-2)!}{(n-1)!} > n^{n-1}.$$

One way to reduce the size of the computations is to apply the theory of superalgebras [Vau98]. Another technique is to decompose the space of multilinear identities into irreducible representations of the symmetric group S_n . The application of the representation theory of the symmetric group to the theory of polynomial identities for algebras was initiated in 1950 by Malcev and Specht. The computational implementation of these techniques was pioneered by Hentzel in the 1970s [Hen77]; for detailed discussions of recent applications see [HP97] and [BH04]. Another approach has been implemented in the Albert system [Jac03]. In this section, we present three small examples ($n \leq 4$) of computational techniques in nonassociative algebra.

Examples:

1. *The identities of degree 3 satisfied by the division algebra of real octonions.*

There are 12 distinct multilinear monomials of degree 3 for a nonassociative algebra:

$$(xy)z, (xz)y, (yx)z, (yz)x, (zx)y, (zy)x, x(yz), x(zy), y(xz), y(zx), z(xy), z(yx).$$

We create a matrix of size 8×12 and initialize it to zero; the columns correspond to the nonassociative monomials. We use a pseudorandom number generator to produce three octonions x, y, z represented as vectors with respect to the standard basis $1, i, j, k, \ell, m, n, p$. We store the evaluation of monomial j in column j of the matrix. For example, generating random integers from the set $\{-1, 0, 1\}$ using the base 3 expansion of $1/\sqrt{2}$ gives

$$x = [1, -1, 0, -1, -1, 1, 0, 0], \quad y = [-1, 1, 1, 1, 0, 0, 1, 0], \quad z = [-1, 1, 1, 1, 0, 0, 0, -1].$$

Evaluation of the monomials gives the matrix in Table 69.1; its reduced row echelon form appears in Table 69.2. The nullspace contains the identities satisfied by the octonion algebra: the span of the rows of the matrix in Table 69.3. These rows represent the linearizations of the right alternative identity (row 1), the left alternative identity (row 2), and the flexible identity (row 5), together with the assocyclic identities $(x, y, z) = (y, z, x)$ and $(x, y, z) = (z, x, y)$ (rows 3 and 4).

2. *The identities of degree 4 satisfied by the Jordan product $x * y = xy + yx$ in every associative algebra over a field of characteristic 0.*

TABLE 69.1 The octonion evaluation matrix

$\begin{bmatrix} -3 & -9 & -9 & -3 & -3 & -9 & -3 & -9 & -9 & -3 & -3 & -9 \\ 1 & -5 & -3 & 5 & -1 & -1 & -3 & -1 & 1 & 1 & -5 & 3 \\ -1 & 1 & 3 & -5 & 1 & -3 & 1 & -1 & 1 & -3 & 3 & -5 \\ 2 & 2 & -4 & -2 & 0 & -2 & 2 & 2 & -4 & -2 & 0 & -2 \\ 1 & 1 & 1 & 3 & -9 & 3 & 5 & -3 & -3 & 7 & -5 & -1 \\ -3 & -3 & 3 & -1 & 5 & -1 & -3 & -3 & 3 & -1 & 5 & -1 \\ -10 & -2 & 0 & 4 & 2 & 4 & -8 & -4 & -2 & 6 & 4 & 2 \\ 0 & 0 & 0 & 6 & -2 & -2 & -2 & 2 & 2 & 4 & -4 & 0 \end{bmatrix}$
--

TABLE 69.2 The reduced row echelon form of the octonion evaluation matrix

$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 & -1 & 1 & 1 & -1 \end{bmatrix}$
--

TABLE 69.3 A basis for the nullspace of the octonion evaluation matrix

$\begin{bmatrix} -1 & -1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & 0 & -1 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

The operation $x*y$ satisfies commutativity in degree 2, and there are no new identities of degree 3, so we consider degree 4. There are 15 distinct multilinear monomials for a commutative nonassociative operation, 12 for association type $((--)--)-$ and 3 for association type $(--)(--)$:

$$\begin{aligned} &((w*x)*y)*z, ((w*x)*z)*y, ((w*y)*x)*z, ((w*y)*z)*x, ((w*z)*x)*y, \\ &((w*z)*y)*x, ((x*y)*w)*z, ((x*y)*z)*w, ((x*z)*w)*y, ((x*z)*y)*w, \\ &((y*z)*x)*w, ((y*z)*w)*x, (w*x)*(y*z), (w*y)*(x*z), (w*z)*(x*y). \end{aligned}$$

When each of these monomials is expanded in terms of the associative product, there are 24 possible terms, namely the permutations of w, x, y, z in lexicographical order: $wxyz, \dots, zyxw$. We construct a 24×15 matrix in which the i, j entry is the coefficient of the i th associative monomial in the expansion of the j th commutative monomial (see Table 69.4). The nontrivial identities of degree 4 satisfied by $x*y$ correspond to the nonzero vectors in the nullspace. The reduced row echelon form appears in Table 69.5. The rank is 11 and, so, the nullspace has dimension 4. A basis for the nullspace consists of the rows of Table 69.6. The first row represents the linearization of the Jordan identity; this is the only identity that involves monomials of both association types. (This proves that the plus algebra A^+ of any associative algebra A is a Jordan algebra.) The Jordan identity implies the identities in the other three rows, which are permuted forms of the identity

$$w*(x*(yz)) - x*(w*(yz)) = (w*(x*y))*z - (x*(w*y))*z + y*(w*(x*z)) - y*(x*(w*z));$$

that is, the commutator of multiplication operators is a derivation.

TABLE 69.4 The Jordan expansion matrix in degree 4

1	0	0	0	0	0	1	1	0	0	0	1	1	0	0
0	1	0	0	0	0	0	0	1	1	0	1	1	0	0
0	0	1	0	0	0	1	1	0	1	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1	1	0	1	0
0	0	0	0	1	0	0	1	1	1	0	0	0	0	1
0	0	0	0	0	1	0	1	0	0	1	1	0	0	1
1	0	1	1	0	0	0	0	0	0	1	0	1	0	0
0	1	0	0	1	1	0	0	0	0	1	0	1	0	0
0	0	1	1	0	1	1	0	0	0	0	0	0	0	1
0	0	0	0	0	1	0	1	0	0	1	1	0	0	1
0	0	0	1	1	1	0	0	1	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1	1	0	1	0
1	1	1	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	1	1	1	0	0	1	0	0	0	0	1	0
1	1	0	0	1	0	1	0	0	0	0	0	0	0	1
0	0	0	0	1	0	0	1	1	1	0	0	0	0	1
0	1	0	0	1	1	0	0	0	0	1	0	1	0	0
0	1	0	0	0	0	0	0	1	1	0	1	1	0	0
1	1	0	0	1	0	1	0	0	0	0	0	0	0	1
0	0	1	1	0	1	1	0	0	0	0	0	0	0	1
1	1	1	0	0	0	0	0	1	0	0	0	0	0	1
0	0	1	0	0	0	1	1	0	1	0	0	0	1	0
1	0	1	1	0	0	0	0	0	0	1	0	1	0	0
1	0	0	0	0	0	1	1	0	0	0	1	1	0	0

TABLE 69.5 The reduced row echelon form of the Jordan expansion matrix

1	0	0	0	0	0	0	0	0	0	-1	0	0	0	0
0	1	0	0	0	0	0	0	0	1	1	1	0	0	1
0	0	1	0	0	0	0	0	0	0	0	-1	0	0	0
0	0	0	1	0	0	0	0	0	1	1	1	0	0	1
0	0	0	0	1	0	0	0	0	0	0	-1	0	0	0
0	0	0	0	0	1	0	0	0	-1	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	-1	0	0	0	0
0	0	0	0	0	0	0	1	0	1	1	1	0	0	1
0	0	0	0	0	0	0	0	1	0	-1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	1	0	-1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	-1

TABLE 69.6 A basis for the nullspace of the Jordan expansion matrix

0	-1	0	-1	0	0	0	-1	0	0	0	0	1	1	1
0	-1	0	-1	0	0	1	-1	1	0	1	0	0	0	0
0	-1	1	-1	1	0	0	-1	0	0	0	1	0	0	0
1	-1	0	-1	0	1	0	-1	0	1	0	0	0	0	0

3. The identities of degree 4 satisfied by the commutator $[x, y] = xy - yx$ in every alternative algebra over a field of characteristic zero.

The group algebra $\mathbb{Q}S_n$ decomposes as a direct sum of full matrix algebras of size $d_\lambda \times d_\lambda$ where the index λ runs over all partitions λ of the integer n ; here d_λ is the dimension of the irreducible representation of S_n corresponding to the partition λ . We choose the “natural representation” to fix a particular decomposition. For each λ there is a projection p_λ from $\mathbb{Q}S_n$ onto the matrix algebra of size $d_\lambda \times d_\lambda$. In the case $n = 4$ the partitions and the dimensions of the corresponding irreducible

TABLE 69.7 Partitions of 4 and irreducible representations of S_4

λ	4	31	22	211	1111
d_λ	1	3	2	3	1

representation S_4 are given in Table 69.7. For a nonassociative operation in degree 4 there are 5 association types:

$$((--)--)-, \quad (-(---))-, \quad (---)(--), \quad -((---)-), \quad -(-(---)),$$

and so any nonassociative identity can be represented as an element of the direct sum of 5 copies of $\mathbb{Q}S_n$: Given a partition λ , the nonassociative identity projects via p_λ to a matrix of size $d_\lambda \times 5d_\lambda$. For an anticommutative operation in degree 4 there are 2 association types:

$$[[[-, -], -], -], \quad [[-, -], [-, -]],$$

and so any anticommutative identity projects via p_λ to a matrix of size $d_\lambda \times 2d_\lambda$. The linearizations of the left and right alternative identities are

$$L(x, y, z) = (xy)z - x(yz) + (yx)z - y(xz), \quad R(x, y, z) = (xy)z - x(yz) + (xz)y - x(z y).$$

Each of these can be “lifted” to degree 4 in five ways; for $L(a, b, c)$ we have

$$wL(x, y, z), \quad L(xw, y, z), \quad L(x, yw, z), \quad L(x, y, zw), \quad L(x, y, z)w;$$

and similarly for $R(x, y, z)$. Altogether we have 10 lifted alternative identities that project via p_λ to a matrix of size $10d_\lambda \times 5d_\lambda$. Using the commutator to expand the two anticommutative association types gives

$$\begin{aligned} [[x, y], z], w] &= ((xy)z)w - ((yx)z)w - (z(xy))w + (z(yx))w \\ &\quad - w((xy)z) + w((yx)z) + w(z(xy)) - w(z(yx)), \\ [[x, y], [z, w]] &= (xy)(zw) - (yx)(zw) - (xy)(wz) + (yx)(wz) \\ &\quad - (zw)(xy) + (zw)(yx) + (wz)(xy) - (wz)(yx). \end{aligned}$$

Given a partition λ we can store these two relations in a matrix of size $2d_\lambda \times 7d_\lambda$: We use all 7 association types, store the right sides of the relations in the first 5 types, and $-I$ (I is the identity matrix) in type 6 (respectively 7) for the first (respectively second) expansion. For each partition λ , all of this data can be stored in a matrix A_λ of size $12d_\lambda \times 7d_\lambda$, which is schematically displayed in Table 69.8. We compute the reduced row echelon form of A_λ : Let i be the largest number for which rows $1 - i$ of $\text{RREF}(A_\lambda)$ have a nonzero entry in the first 5 association types. Then the remaining rows of $\text{RREF}(A_\lambda)$ have only zero entries in the first 5 types; if one of these

TABLE 69.8 The matrix of Malcev identities for partition λ

$A_\lambda =$			0	0
	Lifted alternative identities		$\vdots$	$\vdots$
			0	0
	Expansion of $[[x, y], z], w]$		$-I$	0
	Expansion of $[[x, y], [z, w]]$		0	$-I$

rows contains nonzero entries in the last 2 types, this row represents an identity that is satisfied by the commutator in every alternative algebra. However, such an identity may be a consequence of the obvious anticommutative identities:

$$\begin{aligned} & [[[x, y], z], w] + [[[y, x], z], w] = 0, \\ & [[x, y], [z, w]] + [[y, x], [z, w]] = 0, \quad [[x, y], [z, w]] + [[z, w], [x, y]] = 0. \end{aligned}$$

These identities are represented by a matrix of size $3d_\lambda \times 7d_\lambda$ in which the first $5d_\lambda$ columns are zero. We need to determine if any of the rows $i + 1$ to $12d_\lambda$ of $\text{RREF}(A_\lambda)$ do not lie in the row space of the matrix of anticommutative identities. If such a row exists, it represents a nontrivial identity satisfied by the commutator in every alternative algebra. For example, consider the partition $\lambda = 22$ ($d_\lambda = 2$). The 24×14 matrix A_λ for this partition appears in Table 69.9, and its reduced row echelon form appears in Table 69.10. The 6×14 matrix representing the anticommutative identities for this partition appears in Table 69.11, and its reduced row echelon form appears in Table 69.12. Comparing the last four rows of Table 69.10 with Table 69.12, we see that there is one new identity for $\lambda = 22$ represented by the third-last row of Table 69.10. Similar computations for the other partitions show that there is one nontrivial identity for partition $\lambda = 211$ and no nontrivial identities for the other partitions. The two identities from partitions 22 and 211 are the irreducible components of the Malcev identity: The submodule generated by the linearization of the Malcev identity (in the S_4 -module of all multilinear anticommutative polynomials of degree 4) is the direct sum of two irreducible submodules corresponding to these two partitions.

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70

Lie Algebras

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A Lie algebra is a (nonassociative) algebra satisfying $x^2 = 0$ for all elements x of the algebra (which implies anticommutativity) and the Jacobi identity. Lie algebras arise naturally as (vector) subspaces of associative algebras closed under the commutator operation $[a, b] = ab - ba$. The finite-dimensional simple Lie algebras over algebraically closed fields of characteristic zero occur in many applications. This chapter outlines the structure, classification, and representation theory of these algebras. We also give examples of other types of algebras, e.g., one class of infinite-dimensional simple algebras and one class of finite-dimensional simple Lie algebras over fields of prime characteristic. Section 70.1 is devoted to general definitions about Lie algebras. Section 70.2 discusses semisimple and simple algebras. This section includes the classification of finite-dimensional simple Lie algebras over algebraically closed fields of characteristic zero. Section 70.3 discusses module theory and includes the classification of finite-dimensional irreducible modules for the aforementioned algebras as well as the explicit construction of some of these modules. Section 70.4 discusses graded algebras and modules and uses this formalism to present results on dimensions of irreducible modules.

70.1 Basic Concepts

Unless specified otherwise, F denotes an arbitrary field. All vector spaces and algebras are over F . The reader is referred to Chapter 69 for definitions of many basic algebra terms.

Definitions:

Let A be an algebra over a field F . An **automorphism** is an algebra isomorphism of A to itself. The set of all automorphisms of A is denoted $\text{Aut}(A)$.

A linear transformation $D : A \rightarrow A$ is a **derivation** of A if $D(ab) = D(a)b + aD(b)$ for all $a, b \in A$. The set of all derivations of A is denoted $\text{Der}(A)$.

Let A be an associative algebra and X be a subset of A . Then the smallest ideal of A containing X is denoted by $\langle X \rangle$ and called the **ideal generated by** X .

Let V be a vector space over a field F . Let $V^{\otimes n}$ denote the tensor product of n copies of V and set $T(V) = F1 + \sum_{n=1}^{\infty} V^{\otimes n}$. Define a linear map $T(V) \otimes T(V) \rightarrow T(V)$, $u \otimes v \mapsto uv$, by $1u = u1 = u$ for all $u \in T(V)$ and

$$(v_1 \otimes \dots \otimes v_m)(u_1 \otimes \dots \otimes u_n) = v_1 \otimes \dots \otimes v_m \otimes u_1 \otimes \dots \otimes u_n$$

whenever $m, n \geq 1, v_1, \dots, v_m, u_1, \dots, u_n \in V$. $T(V)$ is the **tensor algebra** on the vector space V . This algebra is defined, and denoted $\bigotimes V$, in Section 13.9.

An algebra L over a field F with product $[\cdot, \cdot] : L \times L \rightarrow L, (a, b) \mapsto [a, b]$ is a **Lie algebra** if it satisfies both

$$[a, a] = 0$$

and

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0 \quad (\text{Jacobi identity})$$

for all $a, b, c \in L$. The first condition implies

$$[a, b] = -[b, a] \quad (\text{anticommutativity})$$

and is equivalent to anticommutativity if the characteristic of $F \neq 2$.

A Lie algebra L is **abelian** if $[a, b] = 0$ for all $a, b \in L$.

If A is an algebra, the vector space A together with the product $[\cdot, \cdot] : A \times A \rightarrow A$ defined by $[a, b] = ab - ba$ is an algebra denoted by A^- .

Let L be a Lie algebra. Let I denote the ideal in $T(L)$ (the tensor algebra on the vector space L) generated by $\{a \otimes b - b \otimes a - [a, b] \mid a, b \in L\}$. The quotient algebra $T(L)/I$ is called the **universal enveloping algebra** of L and is denoted by $U(L)$.

Let V be a vector space with basis X . Define a map $\iota : X \rightarrow T(V)^-$ by $\iota : x \mapsto x \in V^{\otimes 1} \subseteq F1 + \sum_{n=1}^{\infty} V^{\otimes n} = T(V)^-$. Let $Fr(X)$ be the Lie subalgebra of $T(V)^-$ generated by $\iota(X)$. $Fr(X)$ is called the **free Lie algebra generated by X** .

Let V be a vector space and let I be the ideal in $T(V)$ generated by $\{a \otimes b - b \otimes a \mid a, b \in V\}$. The quotient $T(V)/I$ is called the **symmetric algebra** on V and denoted by $S(V)$. The image of $V^{\otimes n}$ in $S(V)$ is denoted by $S^n(V)$. An equivalent construction of this algebra (as a subalgebra, denoted $\bigvee V$, of $T(V)$) is given in Section 13.9.

Let V be a vector space and let I be the ideal in $T(V)$ generated by $\{a \otimes a \mid a \in V\}$. The quotient $T(V)/I$ is called the **exterior algebra** on V and denoted by $\Lambda(V)$. The image of $a_1 \otimes \dots \otimes a_l$ is denoted by $a_1 \wedge \dots \wedge a_l$ and the image of $V^{\otimes n}$ in $\Lambda(V)$ is denoted by $\Lambda^n(V)$. An equivalent construction of this algebra (as a subalgebra, denoted $\bigwedge V$, of $T(V)$) is given in Section 13.9.

Let $End V$ denote the vector space of all linear transformations from V to V (also denoted $L(V, V)$ elsewhere in this book). Let L be a Lie algebra. If $a \in A$, the map $ad : L \rightarrow End L$ defined by $ad(a) : b \mapsto [a, b]$ for all $b \in L$ is called the **adjoint map**.

Let F be a field of prime characteristic p and let L be a Lie algebra over F . If for every $a \in L$ there is some element $a^{[p]} \in L$ such that $(ad(a))^p = ad(a^{[p]})$, then L is called a **p-Lie algebra**.

Facts:

The following facts (except those with a specific reference) can be found in [Jac62, Chap. 5].

1. [Jac62, p. 6] If A is an associative algebra, then A^- is a Lie algebra. If F is a field of prime characteristic p , then A^- is a p -Lie algebra.
2. Let A be an algebra over F . Then $Der(A)$ is a Lie algebra. If F is a field of prime characteristic p , then $Der(A)$ is a p -Lie algebra.
3. Let V be a vector space. The tensor algebra $T(V)$ has the structure of an associative algebra and $T(V)^-$ is a Lie algebra.
4. Let L be a Lie algebra. A subspace $I \subseteq L$ is an *ideal* of L if $[a, b] \in I$ whenever $a \in A, b \in I$. The quotient space L/I with the product $[a + I, b + I] = [a, b] + I$ is a Lie algebra.
5. *Universal property of $U(L)$* : Let L be a Lie algebra. Define a map $\iota : L \rightarrow U(L)$ by $\iota : a \mapsto a \in L^{\otimes 1} \subseteq F1 + \sum_{n=1}^{\infty} L^{\otimes n} = T(L) \rightarrow U(L)$. Then ι is a (Lie algebra) homomorphism of L into

- $U(L)^-$. If A is any associative algebra with unit 1 and ϕ is a homomorphism of L into A^- , then there is a unique homomorphism $\psi : U(L) \rightarrow A$ such that $\psi(1) = 1$ and $\phi = \psi\iota$.
6. *Poincaré–Birkhoff–Witt Theorem*: Let L be a Lie algebra with ordered basis $\{l_i | i \in I\}$. Then $\{l_{i_1} \dots l_{i_k} | k \geq 0, i_1 \leq \dots \leq i_k\}$ is a basis for $U(L)$. Consequently, $\iota : L \rightarrow U(L)$ is injective.
 7. *Universal property of the free Lie algebra*: If L is any Lie algebra and if $\phi : X \rightarrow L$ is any map, there is a unique homomorphism of Lie algebras $\psi : Fr(X) \rightarrow L$ such that $\phi = \psi\iota$.
 8. *Universal property of $S(V)$* : Let V be a vector space. Define a map $\iota : V \rightarrow S(V)$ by $\iota : a \mapsto a \in V^{\otimes 1} \subseteq F1 + \sum_{n=1}^{\infty} V^{\otimes n} \rightarrow S(V)$. Let A be a commutative associative algebra and $\phi : V \rightarrow A$ be a linear map. Then there is a unique algebra homomorphism $\psi : S(V) \rightarrow A$ such that $\phi = \psi\iota$.
 9. *Structure of $S(V)$* : Let V be a vector space with ordered basis $B = \{b_i | i \in I\}$. Then $S(V)$ has basis $\{b_{i_1} \dots b_{i_l} | l \geq 0, i_1 \leq \dots \leq i_l\}$. Consequently, $S(V)$ is isomorphic to the algebra of polynomials on B .
 10. [Lam01, p. 12] *Structure of $\Lambda(V)$* : Let V be a vector space with ordered basis $B = \{b_i | i \in I\}$. Then $\Lambda(V)$ has basis $\{b_{i_1} \wedge \dots \wedge b_{i_l} | l \geq 0, i_1 < \dots < i_l\}$. Consequently, if V has finite dimension l , then $\dim \Lambda(V) = 2^l$.

Examples:

1. $(\text{End } V)^-$ is a Lie algebra, denoted $gl(V)$. Similarly, $(F^{n \times n})^-$ is a Lie algebra, denoted $gl(n, F)$. These algebras are isomorphic.
2. $sl(n, F) = \{x \in gl(n, F) | tr(x) = 0\}$ is a Lie subalgebra, and in fact, a Lie ideal of $gl(n, F)$.
3. Let L be a vector space with basis $\{e, f, h\}$. Defining $[e, f] = h, [h, e] = 2e$, and $[h, f] = -2f$ gives L the structure of a Lie algebra. The linear map $L \rightarrow sl(2, F)$ defined by $e \mapsto \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, f \mapsto \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, h \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is an isomorphism of Lie algebras.

70.2 Semisimple and Simple Algebras

Definitions:

Let L be a Lie algebra. The subspace spanned by all products $[a, b], a, b \in L$, is a subalgebra of L . It is called the **derived algebra** of L and is denoted by $L^{(1)}$.

Let L be a Lie algebra. For $n \geq 2$, $L^{(n)}$ is defined to be $(L^{(n-1)})^{(1)}$ and is called the **n th-derived algebra** of L .

A Lie algebra L is **solvable** if $L^{(n)} = \{0\}$ for some $n \geq 1$.

Let L be a Lie algebra. The sum of all the solvable ideals of L is the **radical** of L and denoted $Rad(L)$. (In Section 69.2, $Rad(L)$ is called the *solvable radical* and denoted $Solv L$.)

A Lie algebra L is **semisimple** if $Rad(L) = \{0\}$. N.B. This is standard terminology in the study of Lie algebras, but does *not* always coincide with the definition of semisimple given for nonassociative algebras in Section 69.2; cf. Fact 5 and Example 3.

A Lie algebra L is **simple** if L contains no nonzero proper ideals and $L^{(1)} \neq \{0\}$. (The second condition excludes the one-dimensional algebra.)

Let A, B be Lie algebras. Then the vector space $A \oplus B$ can be given the structure of a Lie algebra, called the **direct sum** of A and B and also denoted $A \oplus B$, by setting $[a_1 + b_1, a_2 + b_2] = [a_1, a_2] + [b_1, b_2]$ for $a_1, a_2 \in A, b_1, b_2 \in B$.

Let L be a finite-dimensional Lie algebra. The **Killing form**, κ_L , is the symmetric bilinear form on L defined by $\kappa_L(a, b) = tr((ad(a))(ad(b)))$.

For V a finite dimensional vector space over an algebraically closed field F and $x \in \text{End } V$, x is **semisimple** if the minimum polynomial of x has no repeated roots.

Let L be a Lie algebra over an algebraically closed field F and $x \in L$. x is **ad-nilpotent** if $ad(x)$ is a nilpotent linear transformation of L and x is **ad-semisimple** if $ad(x)$ is a semisimple linear transformation of L .

Let L be a Lie algebra. A subalgebra $T \subseteq L$ is a **torus** if every element of T is ad-semisimple.

Let T be a torus in L and let $\alpha \in T^*$, the dual of T . Define L_α , the **α -root space** of L by

$$L_\alpha = \{x \in L \mid [t, x] = \alpha(t)x \ \forall t \in T\}.$$

A vector space E over $\mathbb{R}$ with an inner product (i.e., a positive definite symmetric bilinear form) $\langle \cdot, \cdot \rangle$ is a **Euclidean space**.

Let E be a Euclidean space.

For $0 \neq x \in E$ and $y \in E$ set $\langle y, x \rangle = \frac{2\langle y, x \rangle}{\langle x, x \rangle}$.

For $0 \neq x \in E$ define σ_x , the **reflection** in the hyperplane orthogonal to x , by

$$\sigma_x(y) = y - \langle y, x \rangle x$$

for all $y \in E$.

A finite subset $R \subseteq E$ that spans E and does not contain 0 is a **root system** in E if the following three conditions are satisfied:

- If $x \in R$, $a \in \mathbb{R}$, and $ax \in R$, then $a = \pm 1$.
- If $x \in R$, then $\sigma_x R = R$
- If $x, y \in R$, then $\langle x, y \rangle \in \mathbb{Z}$.

Let R be a root system in E . The **rank** of R is $\dim E$.

A root system R is **decomposable** if $R = R_1 \cup R_2$ with $\emptyset \neq R_1, R_2 \subset R$ and $(R_1, R_2) = \{0\}$. If R is not decomposable, it is **indecomposable**.

Let R be a root system in E . The subgroup of $\text{End } E$ generated by $\{\sigma_\alpha \mid \alpha \in R\}$ is called the **Weyl group** of R .

Let R be a root system in E . A subset $B \subseteq R$ is a **base** for R if B is a basis for E and every $x \in R$ may be written

$$x = \sum_{b \in B} k_b b$$

where all $k_b \geq 0$ or all $k_b \leq 0$.

Let R be a root system in E with base B and let $\alpha \in R$. α is a **positive root** if $\alpha = \sum_{b \in B} k_b b$ where all $k_b \geq 0$. Denote the set of positive roots by R^+ .

Let $B = \{b_1, \dots, b_l\}$ be a base for a root system R . The matrix $[\langle b_i, b_j \rangle]$, is called the **Cartan matrix** of R with respect to B .

Facts:

Most of the following facts (except those with a specific reference) can be found in [Hum72, pp. 35–65].

1. The **radical**, $\text{Rad}(L)$, is a solvable ideal of L .
2. For V a finite dimensional vector space over an algebraically closed field F , $x \in \text{End } V$ is semisimple if and only if x is similar to a diagonal matrix.
3. [Jac62, p. 69] *Cartan's Criterion for Semisimplicity*: Let L be a finite-dimensional Lie algebra over a field of characteristic 0. Then L is semisimple if and only if κ_L is nondegenerate.
4. [Jac62, p. 74] Let L be a finite-dimensional semisimple Lie algebra over a field of characteristic zero and let $D \in \text{Der}(L)$. Then $D = ad(a)$ for some $a \in L$.
5. [Jac62, p. 71] Let L be a finite-dimensional semisimple Lie algebra over a field of characteristic 0. Then L is a direct sum of simple ideals.

6. Let L be a finite-dimensional Lie algebra over an algebraically closed field of characteristic 0. Any torus of L is abelian.
7. Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field of characteristic 0. Let T be a maximal torus of dimension l in L and $\Delta = \{\alpha \in T^* | \alpha \neq 0, L_\alpha \neq \{0\}\}$. Then:
 - $\dim L_\alpha = 1$ for all $\alpha \in \Delta$.
 - $L = T \oplus \sum_{\alpha \in \Delta} L_\alpha$.
 - For $\alpha \in \Delta$ there exists $t_\alpha \in T$ such that $\alpha(t) = \kappa_L(t_\alpha, t)$ for all $t \in T$. Then, defining $(\alpha, \beta) = \kappa_L(t_\alpha, t_\beta)$ gives the $\mathbb{R}$ span of Δ the structure of a Euclidean space of dimension l and Δ is a root system of rank l in this space.
 - Δ is indecomposable if and only if L is simple.
 - Δ spans T^* and so each $w \in W$ acts on T^* .
8. Let R be a root system in E with Weyl group W . Then W is finite and R has a base. Furthermore, if B_1, B_2 are two bases for R , then $B_1 = w(B_2)$ for some $w \in W$. Consequently, when B_1 and B_2 are appropriately ordered, the Cartan matrix of R with respect to B_1 is the same as the Cartan matrix of R with respect to B_2 . Thus, we may refer to the Cartan matrix of R .
9. Let L be a semisimple Lie algebra over a field of characteristic 0, let T_1, T_2 be maximal tori in L , and let Δ_1, Δ_2 be the corresponding root systems. Then there is an automorphism $\phi \in \text{Aut}(L)$ such that $\phi(T_1) = T_2$. Consequently, when bases for Δ_1 and Δ_2 are appropriately ordered, the Cartan matrix for Δ_1 is the same as the Cartan matrix for Δ_2 . Thus, we may refer to the Cartan matrix of L .
10. Let L_1, L_2 be semisimple Lie algebras over an algebraically closed field of characteristic 0. If the Cartan matrices of L_1 and L_2 coincide, then L_1 and L_2 are isomorphic.
11. Let $M = [m_{i,j}]$ be the $l \times l$ Cartan matrix of an indecomposable root system, with base appropriately ordered. Then the diagonal entries of M are all 2 and one of the following occurs.
 - M is of type A_l for some $l \geq 1$: $m_{i,j} = -1$ if $|i - j| = 1$; $m_{i,j} = 0$ if $|i - j| > 1$.
 - M is of type B_l for some $l \geq 3$: $m_{l-1,l} = -2$; $m_{i,j} = -1$ if $|i - j| = 1$ and $(i, j) \neq (l-1, l)$; $m_{i,j} = 0$ if $|i - j| > 1$.
 - M is of type C_l for some $l \geq 2$: $m_{l,l-1} = -2$; $m_{i,j} = -1$ if $|i - j| = 1$ and $(i, j) \neq (l, l-1)$; $m_{i,j} = 0$ if $|i - j| > 1$.
 - M is of type D_l for some $l \geq 4$: $m_{i,j} = -1$ if $|i - j| = 1$ and $(i, j) \neq (l-1, l), (l, l-1)$ or if $(i, j) = (l-2, l), (l, l-2)$; $m_{i,j} = 0$ if $|i - j| > 1$ and $(i, j) \neq (l-2, l), (l, l-2)$ or if $(i, j) = (l-1, l), (l, l-1)$.
 - M is of type E_l , $l = 6, 7, 8$: $m_{i,j} = -1$ if $|i - j| = 1$ and $i, j \neq 2$ or if $(i, j) = (1, 3)(3, 1)(2, 4), (4, 2)$; $m_{i,j} = 0$ if $|i - j| > 1, (i, j) \neq (1, 3)(3, 1)(2, 4), (4, 2)$ or if $|i - j| = 1$ and $i = 2$ or $j = 2$.
 - M is of type F_4 : $M = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -2 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$.
 - M is of type G_2 : $M = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$.
12. Let L be a finite-dimensional simple Lie algebra over an algebraically closed field of characteristic zero. Then L is determined up to isomorphism by its root system with respect to any maximal torus. The Cartan matrix of the root system is of type $A_l, l \geq 1; B_l, l \geq 3; C_l, l \geq 2; D_l, l \geq 4; E_6, E_7, E_8, F_4$, or G_2 .

Examples:

1. The set $\mathbb{R}^3$ with $[\mathbf{u}, \mathbf{v}] = \mathbf{u} \times \mathbf{v}$ (vector cross product) is a three dimensional simple Lie algebra.
2. The set of all upper triangular complex $n \times n$ matrices is a solvable subalgebra of $gl(n, \mathbb{C})$.
3. [Pol69, p. 72] Let $p > 3$ be a prime. The only ideals of $gl(p, \mathbb{Z}_p)$ are $0 \subset scal(p, \mathbb{Z}_p) \subset sl(p, \mathbb{Z}_p) \subset gl(p, \mathbb{Z}_p)$, where $scal(p, \mathbb{Z}_p)$ is the set of scalar matrices. Thus, the only ideals of $L = gl(p, \mathbb{Z}_p)/scal(p, \mathbb{Z}_p)$ are L , $S = sl(p, \mathbb{Z}_p)/scal(p, \mathbb{Z}_p)$ and $\{0\}$. By considering $D = diag(0, 1, 2, \dots, p-1)$, we see that $[sl(n, \mathbb{Z}_p), sl(n, \mathbb{Z}_p)] = sl(n, \mathbb{Z}_p)$, so $[S, S] = S$. Thus, S is not solvable and $Rad(L) = 0$. But L cannot be the sum of simple ideals.
4. The following are Cartan matrices of type A_3, B_3, C_3 respectively:

$$\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -2 \\ 0 & -1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -2 & 2 \end{bmatrix}.$$

5. If $n > 1$ and if n is not a multiple of the characteristic of F , let $L = sl(n, F)$. Then L is a simple Lie algebra of dimension $n^2 - 1$. If T denotes the set of diagonal matrices in $sl(n, F)$ and if $\epsilon_i \in T^*$ is defined by $\epsilon_i(diag(d_1, \dots, d_n)) = d_i$, then T is a maximal torus in $sl(n, F)$, Δ , the set of roots of $sl(n, F)$ with respect to T , is $\{\epsilon_i - \epsilon_j | i \neq j\}$, and the root space $L_{\epsilon_i - \epsilon_j} = FE_{i,j}$. In addition, $\{\epsilon_i - \epsilon_{i+1} | 1 \leq i \leq n-1\}$ is a base for Δ and so the Cartan matrix of $sl(n, F)$ is of type A_{n-1} .

6. Let $(.,.)$ be the symmetric bilinear form on F^{2l+1} with matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & I_l \\ 0 & I_l & 0 \end{bmatrix}$. Then $L = \{x \in gl(2l+1, F) | (xu, v) = -(u, xv) \forall u, v \in F^{2l+1}\}$ is a Lie subalgebra of $gl(2l+1, F)$, denoted by $o(2l+1, F)$. If the characteristic of F is not 2, it is a simple algebra of dimension $2l^2 + l$. Let T denote the set of diagonal matrices in $o(2l+1, F)$. If $\epsilon_i \in T^*$ is defined by $\epsilon_i(diag(d_1, \dots, d_{2l+1})) = d_i$, and if $v_i = \epsilon_{i+1}$ for $1 \leq i \leq l$, then T is a maximal torus in $o(2l+1, F)$; Δ , the set of roots of $o(2l+1, F)$ with respect to T , is $\{\pm v_i | 1 \leq i \leq l\} \cup \{\pm v_i \pm v_j | 1 \leq i \neq j \leq l\}$, and, for $1 \leq i \neq j \leq l$, $L_{v_i} = F(E_{1,l+i+1} - E_{i+1,1})$, $L_{-v_i} = F(E_{1,i+1} - E_{l+i+1,1})$, $L_{v_i - v_j} = F(E_{i+1,j+1} - E_{l+j+1,l+i+1})$, $L_{v_i + v_j} = F(E_{i+1,l+j+1} - E_{j+1,l+i+1})$, $L_{-v_i - v_j} = F(E_{l+i+1,j+1} - E_{l+j+1,i+1})$. In addition, $\{v_i - v_{i+1} | 1 \leq i \leq l-1\} \cup \{v_l\}$ is a base for Δ , and so the Cartan matrix of $o(2l+1, F)$ is of type B_l .

7. Let $(.,.)$ be the skew-symmetric bilinear form on F^{2l} with matrix $\begin{bmatrix} 0 & I_l \\ -I_l & 0 \end{bmatrix}$. Then $L = \{x \in gl(2l, F) | (xu, v) = -(u, xv) \forall u, v \in F^{2l}\}$ is a Lie subalgebra of $gl(2l, F)$, denoted by $sp(2l, F)$. If the characteristic of F is not 2, it is a simple algebra of dimension $2l^2 + l$. Let T denote the set of diagonal matrices in $sp(2l, F)$. If $\epsilon_i \in T^*$ is defined by $\epsilon_i(diag(d_1, \dots, d_{2l})) = d_i$, and if $\mu_i = \epsilon_i$ for $1 \leq i \leq l$, then T is a maximal torus in $sp(2l, F)$, Δ , the set of roots of $sp(2l, F)$ with respect to T , is $\{\pm 2\mu_i | 1 \leq i \leq l\} \cup \{\pm \mu_i \pm \mu_j | i \neq j\}$, and, for $1 \leq i \leq l$, $L_{2\mu_i} = FE_{i,l+i}$, $L_{-2\mu_i} = FE_{l+i,i}$, $L_{\mu_i - \mu_j} = F(E_{i,j} - E_{l+j,l+i})$, $L_{\mu_i + \mu_j} = F(E_{i,l+j} + E_{j,l+i})$, $L_{-\mu_i - \mu_j} = F(E_{l+i,j} + E_{l+j,i})$. In addition, $\{\mu_i - \mu_{i+1} | 1 \leq i \leq l-1\} \cup \{2\mu_l\}$ is a base for Δ , and so the Cartan matrix of $sp(2l, F)$ is of type C_l .

8. Let $(.,.)$ be the symmetric bilinear form on F^{2l} with matrix $\begin{bmatrix} 0 & I_l \\ I_l & 0 \end{bmatrix}$. Then $L = \{x \in gl(2l, F) | (xu, v) = -(u, xv) \forall u, v \in F^{2l}\}$ is a Lie subalgebra of $gl(2l, F)$, denoted by $o(2l, F)$. If the characteristic of F is not 2, it is a simple algebra of dimension $2l^2 - l$. Let T denote the set of diagonal matrices in $o(2l, F)$. If $\epsilon_i \in T^*$ is defined by $\epsilon_i(diag(d_1, \dots, d_{2l})) = d_i$, and if $v_i = \epsilon_i$ for $1 \leq i \leq l$, then T is a maximal torus in $o(2l, F)$, Δ , the set of roots of $o(2l, F)$ with respect to T , is $\{\pm v_i \pm v_j | 1 \leq i \neq j \leq l\}$, and, for $1 \leq i \neq j \leq l$, $L_{v_i - v_j} = F(E_{i,j} - E_{l+j,l+i})$, $L_{v_i + v_j} = F(E_{i,l+j} - E_{j,l+i})$, $L_{-v_i - v_j} = F(E_{l+i,j} - E_{l+j,i})$. In addition, $\{v_i - v_{i+1} | 1 \leq i \leq l-1\} \cup \{v_{l-1} + v_l\}$ is a base for Δ , and so the Cartan matrix of $o(2l, F)$ is of type D_l .

9. Let V be a vector space of dimension $n \geq 1$ over a field of characteristic 0. Let $W(n) = \text{Der}(S(V))$. Then $W(n)$ is an infinite-dimensional simple Lie algebra.
10. Let F be a field of characteristic $p > 0$. Let V be a vector space of dimension $n \geq 1$ with basis $\{x_1, \dots, x_n\}$. Let I denote the ideal $\langle x_1^p, \dots, x_n^p \rangle \subseteq S(V)$, $B(n : 1)$ denote $S(V)/I$, and $W(n : 1) = \text{Der}(B(n : 1))$. Then $W(n : 1)$ is a p -Lie algebra of dimension np^n . It is a simple Lie algebra unless $p = 2$ and $n = 1$.

70.3 Modules

Definitions:

Let A be an associative algebra and V be a vector space over F . A **representation** of A on V is a homomorphism $\phi : A \rightarrow \text{End } V$.

Let L be a Lie algebra and V be a vector space over F . A **representation** of L on V is a homomorphism $\phi : L \rightarrow \text{gl}(V)$.

Let B be an associative algebra or a Lie algebra. A representation $\phi : B \rightarrow \text{gl}(V)$ is **reducible** if there is some nonzero proper subspace $W \subset V$ such that $\phi(x)(W) \subseteq W$ for all $x \in B$. If ϕ is not reducible, it is **irreducible**.

Let L be a Lie algebra and M be a vector space. M is an **L-module** if there is a linear map $L \otimes M \rightarrow M, a \otimes m \mapsto am$ such that $[a, b]m = a(bm) - b(am)$ for all $a, b \in L, m \in M$.

Let M be an L -module. A subspace $N \subseteq M$ is a **submodule** of M if $LN \subseteq N$.

Let M be an L -module. M is **reducible** if M contains a nonzero proper submodule. If M is not reducible it is **irreducible**. If M is a direct sum of irreducible submodules, it is **completely reducible**.

Let M be an L -module and X be a subset of M . The submodule of M **generated by** X is the smallest submodule of M containing X .

Let L be a Lie algebra and M, N be L -modules. A linear transformation $\phi : M \rightarrow N$ is a **homomorphism** of L -modules if $\phi(xm) = x\phi(m)$ for all $x \in L, m \in M$. The set of all L -module homomorphisms from M to N is denoted $\text{Hom}(M, N)$.

Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field of characteristic 0. Let T be a maximal torus in L and M be an L -module. For $\lambda \in T^*$ define M_λ , the λ -**weight space** of M , to be $\{m \in M \mid tm = \lambda(t)m \forall t \in T\}$.

Let L be a finite-dimensional simple Lie algebra over an algebraically closed field of characteristic 0. Let T be a maximal torus, Δ the corresponding root system, B a base for Δ , and Δ^+ the corresponding set of positive roots. Let M be an L -module and $\lambda \in T^*$. An element $0 \neq m \in M_\lambda$ is a **highest weight vector of weight** λ if $L_\alpha m = 0$ for all $\alpha \in \Delta^+$.

Facts:

Unless specified otherwise, V denotes a vector space over a field F .

The following facts may be found in [Hum, Sect. 6].

1. Let L be a Lie algebra and $\phi : L \rightarrow \text{gl}(V)$ be a representation of L on V . Then V may be given the structure of an L -module by setting $xv = \phi(x)(v)$ for all $x \in L, v \in V$. Conversely, if M is an L -module, then the map $\phi : L \rightarrow \text{gl}(M)$ defined by $\phi(x)(m) = xm$ is a representation of L on M . A representation ϕ is irreducible if and only if the corresponding module is.
2. Let ϕ be a representation of a Lie algebra L on V . Then, by the universal property of the universal enveloping algebra, ϕ extends to a representation of $U(L)$ on V . Conversely, every representation of $U(L)$ on V restricts to a representation of L on V . A representation ϕ of $U(L)$ on V is irreducible if and only if its restriction to L is.
3. Let L be a Lie algebra, M be an L -module, and $N \subseteq M$ be a submodule. Then the quotient space M/N may be given the structure of an L -module by setting $x(m + N) = xm + N$ for all $x \in L, m \in M$.

4. Let L be a Lie algebra and M, N be L -modules. Then the vector space $M \oplus N$ may be given the structure of an L -module by setting $x(m + n) = xm + xn$ for all $x \in L, m \in M, n \in N$.
5. Let L be a Lie algebra and M, N be L -modules. Then the vector space $M \otimes N$ may be given the structure of an L -module by setting $x(m \otimes n) = xm \otimes n + m \otimes xn$ for all $x \in L, m \in M, n \in N$.
6. Let L be a Lie algebra and M, N be L -modules. Then $\text{Hom}(M, N)$ may be given the structure of an L -module by setting $(x\phi)(m) = x\phi(m) - \phi(xm)$ for all $x \in L, m \in M$.
7. Let L be a Lie algebra and V be an L -module. Then $T(V)$ is an L -module and the ideals occurring in the definitions of $S(V)$ and $\Lambda(V)$ are submodules. Hence, $S(V)$ and $\Lambda(V)$ are L -modules. Furthermore, each $S^n(V)$ is a submodule of $S(V)$ and each $\Lambda^n(V)$ is a submodule of $\Lambda(V)$.
8. [Jac62, p. 79] *Weyl's Theorem*: Let L be a finite-dimensional semisimple Lie algebra over a field of characteristic zero and let M be a finite-dimensional L -module. Then M is completely reducible.
9. [Hum72, pp. 107–114] Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field of characteristic 0 and M be a finite-dimensional L -module. Let T be a maximal torus in L , Δ be the corresponding root system, $B = \{\alpha_1, \dots, \alpha_l\}$ a base for Δ , and Δ^+ be the corresponding set of positive roots. Then:

- $M = \sum_{\lambda \in T^*} M_\lambda$.
- M contains a highest weight vector of weight λ for some $\lambda \in T^*$ and setting $h_i = \frac{2\alpha_i}{(\alpha_i, \alpha_i)}$ for $1 \leq i \leq l$, we have $\lambda(h_i) \geq 0, \lambda(h_i) \in \mathbb{Z}$ for $1 \leq i \leq l$.
- If M is irreducible and m_1, m_2 are highest weight vectors corresponding to $\lambda_1, \lambda_2 \in T^*$, then $\lambda_1 = \lambda_2$ and $Fm_1 = Fm_2$.
- If M, N are irreducible finite-dimensional L -modules containing highest weight vectors corresponding to the same $\lambda \in T^*$, then M and N are isomorphic.
- Let $\lambda \in T^*$ satisfy $\lambda(h_i) \in \mathbb{Z}, \lambda(h_i) \geq 0$ for $1 \leq i \leq l$. Then there exists a finite-dimensional L -module with highest weight λ .

Examples:

1. Let V be a vector space of dimension $n > 1$. Then V is a $gl(V)$ -module and, hence, a $gl(n, F)$ -module. Therefore, for each $k > 0$, $S^k(V)$ and $\Lambda^k(V)$ are modules for $gl(V)$ and, thus, modules for any subalgebra of $gl(n, F)$.
2. Let V be a vector space with basis $\{x, y\}$. Let $\{e, f, h\}$ be a basis for $sl(2, F)$ with $[e, f] = h, [h, e] = 2e, [h, f] = -2f$. The linear map $sl(2, F) \rightarrow \text{Der}(F[x, y])$ defined by $e \mapsto \frac{x\partial}{\partial y}, f \mapsto \frac{y\partial}{\partial x}, h \mapsto \frac{x\partial}{\partial x} - \frac{y\partial}{\partial y}$ is an isomorphism of $sl(2, F)$ into $\text{Der}(F[x, y])$. Consequently, $F[x, y] = S(V)$ is an $sl(2, F)$ -module and each $S^n(V)$ is an $sl(2, F)$ submodule. $S^n(V)$ has basis $\{x^n, x^{n-1}y, \dots, y^n\}$ and so is an $(n+1)$ -dimensional $sl(2, F)$ -module. It is irreducible.

70.4 Graded Algebras and Modules

Definitions:

Let V be a vector space and A be an additive abelian group. For each $\alpha \in A$, let V_α be a subspace of V . If $V = \bigoplus_{\alpha \in A} V_\alpha$, then V is an **A -graded vector space**.

Let B be an algebra and an A -graded vector space. B is an **A -graded algebra** if $B_\alpha B_\beta \subseteq B_{\alpha+\beta}$ for all $\alpha, \beta \in A$.

Let B be an A -graded associative algebra or an A -graded Lie algebra. Let M be a B -module and an A -graded vector space. M is an **A -graded module** for B if $B_\alpha M_\beta \subseteq M_{\alpha+\beta}$ for all $\alpha, \beta \in A$.

Let V be an A -graded vector space. V has **graded dimension** if $\dim(V_\alpha) < \infty$ for all $\alpha \in A$. In this case, we define the **graded dimension** of V to be the formal sum

$$\text{gr dim}(V) = \sum_{\alpha \in A} \dim(V_\alpha) t^\alpha.$$

The graded dimension of V is sometimes called the **character** of V .

Facts:

- [FLM88, Sect. 1.10] Let V, W be A -graded vector spaces with graded dimensions. Then $V \oplus W$ is a graded vector space with graded dimension and $\text{gr dim}(V \oplus W) = \text{gr dim}(V) + \text{gr dim}(W)$. If W is a subspace of V and $W_\alpha \subseteq V_\alpha$ for all $\alpha \in A$, then the quotient space V/W has graded dimension and $\text{gr dim}(V/W) = \text{gr dim}(V) - \text{gr dim}(W)$. If $\{(\alpha, \beta) \in A \times A \mid V_\alpha, W_\beta \neq \{0\}, \alpha + \beta = \gamma\}$ is finite for all $\gamma \in A$, then $V \otimes W$ is a graded vector space with graded dimension, where $(V \otimes W)_\gamma = \sum_{\alpha+\beta=\gamma} V_\alpha \otimes W_\beta$ and $\text{gr dim}(V \otimes W) = (\text{gr dim}(V))(\text{gr dim}(W))$ (where we set $t^\alpha t^\beta = t^{\alpha+\beta}$).
- [Bou72, p. 36] Let V be a vector space with basis X . Setting $Fr(X)_i = Fr(X) \cap V^{\otimes i}$ gives $Fr(X)$ the structure of a graded Lie algebra. If $|X| = l$ is finite, $Fr(X)$ has graded dimension and $\text{gr dim}(Fr(X)) = \sum_{n>0} \frac{1}{n} (\sum_{d|n} \mu(d) l^{\frac{n}{d}}) t^n$, where μ is the Möbius function (i.e., $\mu(p_1 \dots p_r) = (-1)^r$ if $p_1, \dots, p_r$ are distinct primes and $\mu(n) = 0$ if $p^2 | n$ for some prime p).
- [Jac62, Sect. VIII.3] Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field of characteristic 0, T be a maximal torus in L , Δ be the corresponding root system, W be the Weyl group, Δ^+ be the set of positive roots with respect to some base, and M be a finite-dimensional L -module. Then:
 - Let $\mathbb{Z}\Delta$ denote the additive subgroup of T^* generated by Δ . Then the root space decomposition $L = T + \sum_{\alpha \in \Delta} L_\alpha$ gives L the structure of a $\mathbb{Z}\Delta$ -graded Lie algebra. Here $L_0 = T$.
 - The weight space decomposition $M = \sum_{\alpha \in T^*} M_\alpha$ gives M the structure of a graded module with graded dimension.
 - Weyl character formula:* Assume M is irreducible with highest weight λ . Let $\delta = \frac{1}{2} \sum_{\alpha \in \Delta^+} \alpha$. Then

$$\text{gr dim}(M) = \left(\sum_{w \in W} (\det(w) t^{w(\lambda+\delta)}) \right) / \left(\sum_{w \in W} (\det(w) t^{w\delta}) \right).$$

Examples:

- Setting $T(V)_i = V^{\otimes i}$ gives $T(V)$ the structure of a $\mathbb{Z}$ -graded algebra. If V has finite dimension l , then $T(V)$ has graded dimension and $\text{gr dim}(T(V)) = (1 - lt)^{-1}$.
- Setting $S(V)_i = S^i(V)$ gives $S(V)$ the structure of a $\mathbb{Z}$ -graded algebra. If V has finite dimension l , then $S(V)$ has graded dimension and $\text{gr dim}(S(V)) = (1 - t)^{-l}$.
- Setting $\Lambda(V)_i = \Lambda^i(V)$ gives $\Lambda(V)$ the structure of a $\mathbb{Z}$ -graded algebra. If V has finite dimension l , then $\Lambda(V)$ has graded dimension and $\text{gr dim}(\Lambda(V)) = (1 + t)^l$.
- Let L be a finite-dimensional semisimple Lie algebra over an algebraically closed field of characteristic 0, T be a maximal torus in L , Δ be the corresponding root system, and $B = \{\alpha_1, \dots, \alpha_l\}$ be a base. For $1 \leq i \leq l$ define λ_i by $\lambda_i(\frac{2t_{\alpha_j}}{(\alpha_j, \alpha_j)}) = \delta_{i,j}$. Then:
 - If $L = sl(V)$, where V is an $l + 1$ -dimensional vector space and the base for Δ is as described in Example 5 of section 70.2, then $\Lambda^i(V)$ is the irreducible $sl(V)$ -module of highest weight λ_i for $1 \leq i \leq l$.

- If $L = o(V)$, where V is a $2l + 1$ -dimensional vector space and the base for Δ is as described in Example 6 of section 70.2, then $\Lambda^i(V)$ is the irreducible $o(V)$ -module of highest weight λ_i for $1 \leq i \leq l - 1$.
- If $L = o(V)$, where V is a $2l$ -dimensional vector space and the base for Δ is as described in Example 8 of section 70.2, then $\Lambda^i(V)$ is the irreducible $o(V)$ -module of highest weight λ_i for $1 \leq i \leq l - 2$.

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V

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MATLAB[®] is generally recognized as the leading software for scientific computation. It was originally developed in the 1970s by Cleve Moler as an interactive Matrix Laboratory with matrix routines based on the algorithms in the LINPACK and EISPACK software libraries. In the original 1978 version everything in MATLAB was done with matrices, even the graphics. MATLAB has continued to grow and expand from the classic 1978 Fortran version to the current version, MATLAB 7, which was released in May 2004. Each new release has included significant improvements. The graphics capabilities were greatly enhanced with the introduction of Handle Graphics and Graphical User Interfaces in version 4 (1992). A sparse matrix package was also included in version 4. Over the years, dozens of toolboxes (application libraries of specialized MATLAB files) have been added in areas such as signal processing, statistics, optimization, symbolic math, splines, and image processing. MATLAB's matrix computations are now based on the LAPACK and BLAS software libraries. MATLAB is widely used in linear algebra courses. Books such as [Leo06], [LHF03], and [HZ04] have extensive sets of MATLAB exercises and projects for the standard first course in linear algebra. The book by D.J. Higham and N.J. Higham [HH03] provides a comprehensive guide to all the basic features of MATLAB.

71.1 Matrices, Submatrices, and Multidimensional Arrays

Facts:

1. The basic data elements that MATLAB uses are matrices. Matrices can be entered into a MATLAB session using square brackets.
2. If A and B are matrices with the same number of rows, then one can append the columns of B to the matrix A and form an augmented matrix C by setting $C = [A, B]$. If E is a matrix with the same number of columns as A , then one can append the rows of E to A and form an augmented matrix F by setting $F = [A; E]$.

3. Row vectors of evenly spaced entries can be generated using MATLAB's : operator.
4. A submatrix of a matrix A is specified by $A(u, v)$ where u and v are vectors that specify the row and column indices of the submatrix.
5. MATLAB arrays can have more than two dimensions.

Commands:

1. The number of rows and columns of a matrix can be determined using MATLAB's size command.
2. The command $\text{length}(x)$ can be used to determine the number of entries in a vector x . The length command is equivalent to the command $\max(\text{size}(x))$.
3. MATLAB's cat command can be used to concatenate two or more matrices along a single dimension and it can also be used to create multidimensional arrays. If two matrices A and B have the same number of columns, then the command $\text{cat}(1, A, B)$ produces the same matrix as the command $[A; B]$. If the matrices B and C have the same number of rows, the command $\text{cat}(2, B, C)$ produces the same matrix as the command $[B, C]$. The cat command can be used with more than two matrices as arguments. For example, the command $\text{cat}(2, A, B, C, D)$ generates the same matrix as the command $[A, B, C, D]$. If $A_1, A_2, \dots, A_k$ are $m \times n$ matrices, the command $S = \text{cat}(3, A_1, A_2, \dots, A_k)$ will generate an $m \times n \times k$ array S .

Examples:

1. The matrix

$$A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 3 & 5 & 7 & 2 \\ 2 & 4 & 1 & 1 \end{bmatrix}$$

is generated using the command

$$A = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 3 & 5 & 7 & 2 \\ 2 & 4 & 1 & 1 \end{bmatrix}.$$

Alternatively, one could generate the matrix on a single line using semicolons to designate the ends of the rows

$$A = [1 \ 2 \ 4 \ 3; 3 \ 5 \ 7 \ 2; 2 \ 4 \ 1 \ 1].$$

The command $\text{size}(A)$ will return the vector $(3, 4)$ as the answer. The command $\text{size}(A, 1)$ will return the value 3, the number of rows of A , and the command $\text{size}(A, 2)$ will return the value 4, the number of columns of A .

2. The command

$$x = 3 : 7$$

will generate the vector $x = (3, 4, 5, 6, 7)$. To change the step size to $\frac{1}{2}$, set

$$z = 3 : 0.5 : 7.$$

This will generate the vector $z = (3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7)$. The MATLAB commands $\text{length}(x)$ and $\text{length}(z)$ will generate the answers 5 and 9, respectively.

3. If A is the matrix in Example 1, then the command $A(2,:)$ will generate the second row vector of A and the command $A(:, 3)$ will generate the third column vector of A . The submatrix of elements that are in the first two rows and last two columns is given by $A(1:2, 3:4)$. Actually there is no need to use adjacent rows or columns. The command $C = A([1 \ 3], [1 \ 3 \ 4])$ will generate the submatrix

$$C = \begin{bmatrix} 1 & 4 & 3 \\ 2 & 1 & 1 \end{bmatrix}.$$

4. The command

$$S = \text{cat}(3, [5 \ 1 \ 2; 3 \ 2 \ 1], [1 \ 2 \ 3; 4 \ 5 \ 6])$$

will produce a $2 \times 3 \times 2$ array S with

$$S(:, :, 1) = \begin{bmatrix} 5 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

and

$$S(:, :, 2) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

71.2 Matrix Arithmetic

The six basic MATLAB operators for doing matrix arithmetic are: $+$, $-$, $*$, $^$, $\backslash$, and $/$. The matrix left and right divide operators, $\backslash$ and $/$, are described in Section 71.5. These same operators are also used for doing scalar arithmetic.

Facts:

1. If A and B are matrices with the same dimensions, then their sum and difference are computed using the commands: $A + B$ and $A - B$.
2. If B and C are matrices and the multiplication BC is possible, then the product $E = BC$ is computed using the command

$$E = B * C.$$

3. The k th power of a square matrix A is computed with the command A^k .
4. Scalars can be either real or complex numbers. A complex number such as $3 + 4i$ is entered in MATLAB as $3 + 4i$. It can also be entered as $3 + 4 * \text{sqrt}(-1)$ or by using the command $\text{complex}(3, 4)$. If i is used as a variable and assigned a value, say $i = 5$, then MATLAB will assign the expression $3 + 4 * i$ the value 23; however, the expression $3 + 4i$ will still represent the complex number $3 + 4i$. In the case that i is used as a variable and assigned a numerical value, one should be careful to enter a complex number of the form $a + i$ (where a real) as $a + 1i$.
5. MATLAB will perform arithmetic operations element-wise when the operator is preceded by a period in the MATLAB command.
6. The conjugate transpose of a matrix B is computed using the command B' . If the matrix B is real, then B' will be equal to the transpose of B . If B has complex entries, then one can take its transpose without conjugating using the command B' .

Commands:

1. The inverse of a nonsingular matrix C is computed using the command $\text{inv}(C)$.
2. The determinant of a square matrix A is computed using the command $\text{det}(A)$.

Examples:

1. If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix}$$

the commands $A * B$ and $A \cdot B$ will generate the matrices

$$\begin{bmatrix} 9 & 7 \\ 23 & 15 \end{bmatrix}, \quad \begin{bmatrix} 5 & 2 \\ 6 & 12 \end{bmatrix}.$$

71.3 Built-In MATLAB Functions

The `inv` and `det` commands are examples of built-in MATLAB functions. Both functions have a single input and a single output. Thus, the command `d = det(A)` has the matrix A as its input argument and the scalar d as its output argument. A MATLAB function may have many input and output arguments. When a command of the form

$$[A_1, \dots, A_k] = \text{fname}(B_1, \dots, B_n) \quad (71.1)$$

is used to call a function `fname` with input arguments $B_1, \dots, B_n$, MATLAB will execute the function routine and return the values of the output arguments $A_1, \dots, A_k$.

Facts:

1. The number of allowable input and output arguments for a MATLAB function is defined by a function statement in the MATLAB file that defines the function. (See section 71.8.) The function may require some or all of its input arguments. A MATLAB command of the form (71.1) may be used with j output arguments where $0 \leq j \leq k$. The MATLAB help facility describes the various input and output options for each of the MATLAB commands.

Examples:

1. The MATLAB function `pi` is used to generate the number π . This function is used with no input arguments.
2. The MATLAB function `kron` has two input arguments. If A and B are matrices, then the command `kron(A,B)` computes the Kronecker product of A and B . Thus, if $A = [1, 2; 3, 4]$ and $B = [1, 1; 1, 1]$, then the command `K = kron(A, B)` produces the matrix

$$K = \begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \\ 3 & 3 & 4 & 4 \\ 3 & 3 & 4 & 4 \end{bmatrix}$$

and the command `L = kron(B, A)` produces the matrix

$$L = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 2 & 1 & 2 \\ 3 & 4 & 3 & 4 \end{bmatrix}.$$

- One can compute the QZ factorization (see section 71.6) for the generalized eigenvalue problem using a command

$$[E, F, Q, Z] = \text{qz}(A, B)$$

with two input arguments and four outputs. The input arguments are square matrices A and B and the outputs are quasitriangular matrices E and F and unitary matrices Q and Z such that

$$QAZ = E \quad \text{and} \quad QBZ = F.$$

The command

$$[E, F, Q, Z, V, W] = \text{qz}(A, B)$$

will also compute matrices V and W of generalized eigenvectors.

71.4 Special Matrices

The ELMAT directory of MATLAB contains a collection of MATLAB functions for generating special types of matrices.

Commands:

- The following table lists commands for generating various types of special matrices.

Matrix	Command Syntax	Description
eye	eye(n)	Identity matrix
ones	ones(n) or ones(m, n)	Matrix whose entries are all equal to 1
zeros	zeros(n) or zeros(m, n)	Matrix whose entries are all equal to 0
rand	rand(n) or rand(m, n)	Random matrix
compan	compan(p)	Companion matrix
hadamard	hadamard(n)	Hadamard matrix
gallery	gallery(matname, p1, p2, ...)	Large collection of special test matrices
hankel	hankel(c) or hankel(c, r)	Hankel matrix
hilb	hilb(n)	Hilbert matrix
invhilb	invhilb(n)	Inverse Hilbert matrix
magic	magic(n)	Magic square
pascal	pascal(n)	Pascal matrix
rosser	rosser	Test matrix for eigenvalue solvers
toeplitz	toeplitz(c) or toeplitz(c, r)	Toeplitz matrix
vander	vander(x)	Vandermonde matrix
wilkinson	wilkinson(n)	Wilkinson's eigenvalue test matrix

- The command `gallery` can be used to access a large collection of test matrices developed by N. J. Higham. Enter `help gallery` to obtain a list of all classes of gallery test matrices.

Examples:

- The command `rand( $n$ )` will generate an $n \times n$ matrix whose entries are random numbers that are uniformly distributed in the interval $(0, 1)$. The command may be used with two input arguments

to generate nonsquare matrices. For example, the command `rand(3,2)` will generate a random 3×2 matrix. The command `rand` when used by itself with no input arguments will generate a single random number between 0 and 1.

2. The command

$$A = [\text{eye}(2), \text{ones}(2,3); \text{zeros}(2), 2 * \text{ones}(2,3)]$$

will generate the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 2 & 2 & 2 \end{bmatrix}$$

3. The command `toeplitz(c)` will generate a symmetric toeplitz matrix whose first column is the vector `c`. Thus, the command

$$\text{toeplitz}([1; 2; 3])$$

will generate

$$T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Note that in this case, since the `toeplitz` command was used with no output argument, the computed value of the command `toeplitz(c)` was assigned to the temporary variable `ans`. Further computations may end up overwriting the value of `ans`. To keep the matrix for further use in the MATLAB session, it is advisable to include an output argument in the calling statement.

For a nonsymmetric Toeplitz matrix it is necessary to include a second input argument `r` to define the first row of the matrix. If $r(1) \neq c(1)$, the value of $c(1)$ is used for the main diagonal. Thus, commands

$$c = [1; 2; 3], \quad r = [9, 5, 7], \quad T = \text{toeplitz}(c, r)$$

will generate

$$T = \begin{bmatrix} 1 & 5 & 7 \\ 2 & 1 & 5 \\ 3 & 2 & 1 \end{bmatrix}$$

The Toeplitz matrix generated is stored using the variable `T`.

4. One of the classes of gallery test matrices is circulant matrices. These are generated using the MATLAB function `circul`. To see how to use this function, enter the command

$$\text{help private}\backslash\text{circul}.$$

The help information will tell you that the `circul` function requires an input vector `v` and that the command

$$C = \text{gallery}('circul', v)$$

will generate a circulant matrix whose first row is $\mathbf{v}$. Thus, the command

$$\mathbf{C} = \text{gallery}('circul', [4, 5, 6])$$

will generate the matrix

$$\mathbf{C} = \begin{bmatrix} 4 & 5 & 6 \\ 6 & 4 & 5 \\ 5 & 6 & 4 \end{bmatrix}.$$

71.5 Linear Systems and Least Squares

The simplest way to solve a linear system in MATLAB is to use the matrix left divide operator.

Facts:

1. The symbol $\backslash$ represents MATLAB's matrix left divide operator. One can compute the solution to a linear system $\mathbf{Ax} = \mathbf{b}$ by setting

$$\mathbf{x} = \mathbf{A} \backslash \mathbf{b}.$$

If $\mathbf{A}$ is an $n \times n$ matrix, then MATLAB will compute the solution using Gaussian elimination with partial pivoting. A warning message is given when the matrix is badly scaled or nearly singular. If the coefficient matrix is nonsquare, then MATLAB will return a least squares solution to the system that is essentially equivalent to computing $\mathbf{A}^\dagger \mathbf{b}$ (where $\mathbf{A}^\dagger$ denotes the pseudoinverse of $\mathbf{A}$). In this case, MATLAB determines the numerical rank of the coefficient matrix using a QR decomposition and gives a warning when the matrix is rank deficient.

If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is $m \times k$, then the command

$$\mathbf{C} = \mathbf{A} \backslash \mathbf{B}$$

will produce an $n \times k$ matrix whose column vectors satisfy

$$\mathbf{c}_j = \mathbf{A} \backslash \mathbf{b}_j \quad j = 1, \dots, k.$$

2. The symbol $/$ represents MATLAB's matrix right divide operator. It is defined by

$$\mathbf{B}/\mathbf{A} = (\mathbf{A}' \backslash \mathbf{B}')'.$$

In the case that $\mathbf{A}$ is nonsingular, the computation $\mathbf{B}/\mathbf{A}$ is essentially the same as computing $\mathbf{B}\mathbf{A}^{-1}$, however, the computation is carried out without actually computing $\mathbf{A}^{-1}$.

Commands:

The following table lists some of the main MATLAB commands that are useful for linear systems.

Function	Command Syntax	Description
rref	<code>U = rref(A)</code>	Reduced row echelon form of a matrix
lu	<code>[L , U] = lu(A)</code>	LU factorization
linsolve	<code>x = linsolve(A , b , opts)</code>	Efficient solver for structured linear systems
chol	<code>R = chol(A)</code>	Cholesky factorization of a matrix
norm	<code>p = norm(X)</code>	Norm of a matrix or a vector
null	<code>U = null(A)</code>	Basis for the null space of a matrix
null	<code>R = null(A, 'r')</code>	Basis for null space rational form
orth	<code>Q = orth(A)</code>	Orthonormal basis for the column space of a matrix
rank	<code>r = rank(A)</code>	Numerical rank of a matrix
cond	<code>c = cond(A)</code>	2-norm condition number for solving linear systems
rcond	<code>c = rcond(A)</code>	Reciprocal of approximate 1-norm condition number
qr	<code>[Q , R] = qr(A)</code>	QR factorization
svd	<code>s = svd(A)</code>	Singular values of a matrix
svd	<code>[U , S , V] = svd(A)</code>	Singular value decomposition
pinv	<code>B = pinv(A)</code>	Pseudoinverse of a matrix

Examples:

1. The null command can be used to produce an orthonormal basis for the nullspace of a matrix. It can also be used to produce a “rational” nullspace basis obtained from the reduced row echelon form of the matrix. If

$$A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix},$$

then the command `U = null(A)` will produce the matrix

$$U = \begin{bmatrix} -0.8165 & -0.0000 \\ 0.4082 & 0.7071 \\ 0.4082 & -0.7071 \\ -0.0000 & 0.0000 \end{bmatrix}$$

where the entries of U are shown in MATLAB's `format short` (with four-digit mantissas). The column vectors of U form an orthonormal basis for the nullspace of A . The command `R = null(A, 'r')` will produce a matrix R whose columns form a simple basis for the nullspace.

$$R = \begin{bmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

2. MATLAB defines the numerical rank of a matrix to the number of singular values of the matrix that are greater than

$$\max(\text{size}(A)) * \text{norm}(A) * \text{eps}$$

where `eps` has the value 2^{-52} , which is a measure of the precision used in MATLAB computations. Let H be the 12×12 Hilbert matrix. The singular values of H can be computed using the command `s = svd(H)`. The smallest singular values are $s(11) \approx 2.65 \times 10^{-14}$ and $s(12) \approx 10^{-16}$. Since the value of `eps` is approximately 2.22×10^{-16} , the computed value of `rank(H)` will be the numerical rank 11, even though the exact rank of H is 12. The computed value of `cond(H)` is approximately 1.8×10^{16} and the computed value of `rcond(H)` is approximately 2.6×10^{-17} .

71.6 Eigenvalues and Eigenvectors

MATLAB's `eig` function can be used to compute both the eigenvalues and eigenvectors of a matrix.

Commands:

1. The `eig` command. Given a square matrix A , the command `e = eig(A)` will generate a column vector e whose entries are the eigenvalues of A . The command `[X, D] = eig(A)` will generate a matrix X whose column vectors are the eigenvectors of A and a diagonal matrix D whose diagonal entries are the eigenvalues of A .
2. The `eigshow` command. MATLAB's `eigshow` utility provides a visual demonstration of eigenvalues and eigenvectors of 2×2 matrices. The utility is invoked by the command `eigshow(A)`. The input argument A must be a 2×2 matrix. The command can also be used with no input argument, in which case MATLAB will take $[1 \ 3; 4 \ 2]/4$ as the default 2×2 matrix. The `eigshow` utility shows how the image Ax changes as we rotate a unit vector x around a circle. This rotation is carried out manually using a mouse. If A has real eigenvalues, then we can observe the eigenvectors of the matrix when the vectors x and Ax are in the same or opposite directions.
3. The command `J = jordan(A)` can be used to compute the Jordan canonical form of a matrix A . This command will only give accurate results if the entries of A are exactly represented, i.e., the entries must be integers or ratios of small integers. The command `[X, J] = jordan(A)` will also compute the similarity matrix X so that $A = XJX^{-1}$.
4. The following table lists some additional MATLAB functions that are useful for eigenvalue related problems.

Function	Command Syntax	Description
<code>poly</code>	<code>p = poly(A)</code>	Characteristic polynomial of a matrix
<code>hess</code>	<code>H = hess(A)</code> or <code>[U, H] = hess(A)</code>	Hessenberg form
<code>schur</code>	<code>T = schur(A)</code> or <code>[U, T] = schur(A)</code>	Schur decomposition
<code>qz</code>	<code>[E, F, Q, Z] = qz(A, B)</code>	QZ factorization for generalized eigenvalues
<code>condeig</code>	<code>s = condeig(A)</code>	Condition numbers for the eigenvalues of A
<code>expm</code>	<code>E = expm(A)</code>	Matrix exponential

71.7 Sparse Matrices

A matrix is *sparse* if most of its entries are zero. MATLAB has a special data structure for handling sparse matrices. This structure stores the nonzero entries of a sparse matrix together with their row and column indices.

Commands:

1. The command `sparse` is used to generate sparse matrices. When used with a single input argument the command

$$S = \text{sparse}(A)$$

will convert an ordinary MATLAB matrix A into a matrix S having the sparse data structure. More generally, a command of the form

$$S = \text{sparse}(i, j, s, m, n, \text{nzmax})$$

will generate an $m \times n$ sparse matrix S whose nonzero entries are the entries of the vector s . The row and column indices of the nonzero entries are given by the vectors i and j . The last input argument nzmax specifies the total amount of space allocated for nonzero entries. If the allocation argument is omitted, by default MATLAB will set it to equal the value of $\text{length}(s)$.

- MATLAB's `spy` command can be used to plot the sparsity pattern of a matrix. In these plots the matrix is represented by a rectangular box with dots corresponding to the positions of its nonzero entries.
- The MATLAB directory `SPARFUN` contains a large collection of MATLAB functions for working with sparse matrices. The general sparse linear algebra functions are given in the following table.

MATLAB Function	Description
<code>eigs</code>	A few eigenvalues, using ARPACK
<code>svds</code>	A few singular values, using <code>eigs</code>
<code>luinc</code>	Incomplete LU factorization
<code>cholinc</code>	Incomplete Cholesky factorization
<code>normest</code>	Estimate the matrix 2-norm
<code>condest</code>	1-norm condition number estimate
<code>sprank</code>	Structural rank

All of these functions require a sparse matrix as an input argument. All have one basic output argument except in the case of `luinc`, where the basic output consists of the L and U factors.

- The `SPARFUN` directory also includes a collection of routines for the iterative solution of sparse linear systems.

MATLAB Function	Description
<code>pcg</code>	Preconditioned Conjugate Gradients Method
<code>bicg</code>	BiConjugate Gradients Method
<code>bicgstab</code>	BiConjugate Gradients Stabilized Method
<code>cgs</code>	Conjugate Gradients Squared Method
<code>gmres</code>	Generalized Minimum Residual Method
<code>lsqr</code>	Conjugate Gradients on the Normal Equations
<code>minres</code>	Minimum Residual Method
<code>qmr</code>	Quasi-Minimal Residual Method
<code>symmlq</code>	Symmetric LQ Method

If A is a sparse coefficient matrix and B is a matrix of right-hand sides, then one can solve the equation $AX = B$ using a command of the form $X = \text{fname}(A, B)$, where `fname` is one of the iterative solver functions in the table.

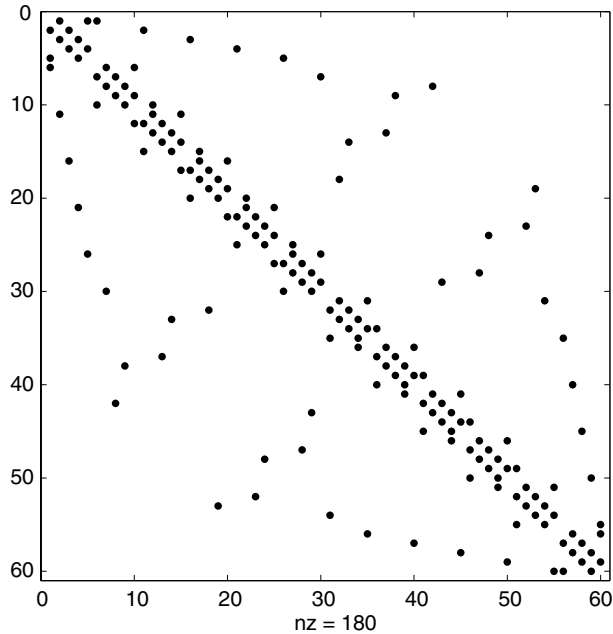
Examples:

- The command

$$S = \text{sparse}([25, 37, 8], [211, 15, 92], [4.5, 3.2, 5.7], 200, 300)$$

will generate a 200×300 sparse matrix S whose only nonzero entries are

$$s_{25,211} = 4.5, \quad s_{37,15} = 3.2, \quad s_{8,92} = 5.7.$$

FIGURE 71.1 `Spy(B)`.

2. The command `B = bucky` will generate the 60×60 sparse adjacency matrix B of the connectivity graph of the Buckminster Fuller geodesic dome and the command `spy(B)` will generate the spy plot shown in Figure 71.1.

71.8 Programming

MATLAB has built in all of the main structures one would expect from a high-level computer language. The user can extend MATLAB by adding on programs and new functions.

Facts:

1. MATLAB programs are called *M-files* and should be saved with a `.m` extension.
2. MATLAB programs may be in the form of *script* files that list a series of commands to be executed when the file is called in a MATLAB session, or they can be in the form of MATLAB functions.
3. MATLAB programs frequently include for loops, while loops, and if statements.
4. A function file must start with a function statement of the form

$$\text{function } [\text{oarg1}, \dots, \text{oargk}] = \text{fname}(\text{iarg1}, \dots, \text{iargj})$$

where `fname` is the name of the function, `iarg1, ..., iargj` are its input arguments, and `oarg1, ..., oargk` are the output arguments. In calling a MATLAB function, it is not necessary to use all of the input and output allowed for in the general syntax of the command. In fact, MATLAB functions are commonly used with no output arguments whatsoever.

5. One can construct simple functions interactively in a MATLAB session using MATLAB's inline command. A simple function such as $f(t) = t^2 + 4$ can be described by the character array (or string)

" $t^2 + 4$." The inline command will transform the string into a function for use in the current MATLAB session. Inline functions are particularly useful for creating functions that are used as input arguments for other MATLAB functions. An inline function is not saved as an m-file and consequently is lost when the MATLAB session is ended.

6. One can use the same MATLAB command with varying amounts of input and output arguments. MATLAB keeps track of the number of input and output arguments included in the call statement using the functions `nargin` (the number of input arguments) and `nargout` (the number of output arguments). These commands are used inside the body of a MATLAB function to tailor the computations and output to the specifications of the calling statement.
7. MATLAB has six relational operators that are used for comparisons of scalars or elementwise comparisons of arrays. These operators are:

Relational Operators	
<	less than
<=	less than or equal
>	greater than
>=	greater than or equal
==	equal
~=	not equal

8. There are three logical operators as shown in the following table:

Logical Operators	
&	AND
	OR
~	NOT

These logical operators regard any nonzero scalar as corresponding to TRUE and 0 as corresponding to FALSE. The operator `&` corresponds to the logical AND. If a and b are scalars, the expression $a \& b$ will equal 1 if a and b are both nonzero (TRUE) and 0 otherwise. The operator `|` corresponds to the logical OR. The expression $a | b$ will have the value 0 if a and b are both 0 and otherwise it will be equal to 1. The operator `~` corresponds to the logical NOT. For a scalar a , the expression $\sim a$ takes on the value 1 (TRUE) if $a = 0$ (FALSE) and the value 0 (FALSE) if $a \neq 0$ (TRUE).

For matrices these operators are applied element-wise. Thus, if A and B are both $m \times n$ matrices, then $A \& B$ is a matrix of zeros and ones whose (i, j) entry is $a(i, j) \& b(i, j)$.

Examples:

1. Given two $m \times n$ matrices A and B , the command $C = A < B$ will generate an $m \times n$ matrix consisting of zeros and ones. The (i, j) entry will be equal to 1 if and only if $a_{ij} < b_{ij}$. If

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 4 & -2 & 5 \\ 1 & -3 & -2 \end{bmatrix},$$

then command $A >= 0$ will generate

$$\begin{aligned} \text{ans} = \\ & \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{aligned}$$

2. If

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix},$$

then

$$A \& B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad A|B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \sim A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

3. To construct the function $g(t) = 3 \cos t - 2 \sin t$ interactively set

$$g = \text{inline}('3 * \cos(t) - 2 * \sin(t)').$$

If one then enters $g(0)$ on the command line, MATLAB will return the answer 3. The command `ezplot(g)` will produce a plot of the graph of $g(t)$. (See Section 71.9 for more information on producing graphics.)

4. If the numerical nullity of a matrix is defined to be the number of columns of the matrix minus the numerical rank of the matrix, then one can create a file *numnull.m* to compute the numerical nullity of a matrix. This can be done using the following lines of code.

```
function k = numnull(A)
% The command numnull(A) computes the numerical nullity of A
[m,n] = size(A);
k = n - rank(A);
```

The line beginning with the % is a comment that is not executed. It will be displayed when the command `help numnull` is executed. The semicolons suppress the printouts of the individual computations that are performed in the function program.

5. The following is an example of a MATLAB function to compute the circle that gives the best least squares fit to a collection of points in the plane.

```
function [center,radius,e] = circfit(x,y,w)
% The command [center,radius] = circfit(x,y) generates
% the center and radius of the circle that gives the
% best least squares fit to the data points specified
% by the input vectors x and y. If a third input
% argument is specified then the circle and data
% points will be plotted. Specify a third output
% argument to get an error vector showing how much
% each point deviates from the circle.
if size(x,1) == 1 & size(y,1) == 1
    x = x'; y = y';
end
A = [2*x, 2*y, ones(size(x))];
b = x.^2 + y.^2;
c = A\b;
center = c(1:2)';
radius = sqrt(c(3) + c(1)^2 + c(2)^2);
```

```

if nargin > 2
    t = 0:0.1:6.3;
    u = c(1) + radius * cos(t);
    v = c(2) + radius * sin(t);
    plot(x,y,'x',u,v)
    axis('equal')
end
if nargin == 3
    e = sqrt((x - c(1)).^2 + (y - c(2)).^2) - radius;
end

```

The command `plot(x,y,'x',u,v)` is used to plot the original (x, y) data as discrete points in the plane, with each point designated by an “x,” and to also, on the same axis system, plot the (u, v) data points as a continuous curve. The following section explains MATLAB plot commands in greater detail.

71.9 Graphics

MATLAB graphics utilities allow the user to do simple two- and three-dimensional plots as well as more sophisticated graphical displays.

Facts:

1. MATLAB incorporates an objected-oriented graphics system called *Handle Graphics*. This system allows the user to modify and add on to existing figures and is useful in producing computer animations.
2. MATLAB's graphics capabilities include digital imaging tools. MATLAB images may be indexed or true color.
3. An indexed image requires two matrices, a $k \times 3$ colormap matrix whose rows are triples of numbers that specify red, green, blue intensities, and an $m \times n$ image matrix whose entries assign a colormap triple to each pixel of the image.
4. A true color image is one derived from an $m \times n \times 3$ array, which specifies the red, green, blue triplets for each pixel of the image.

Commands:

1. The `plot` command is used for simple plots of x - y data sets. Given a set of (x_i, y_i) data points, the command `plot(x,y)` plots the data points and by default sequentially draws line segments to connect the points. A third input argument may be used to specify a color (the default color for plots is black) or to specify a different form of plot such as discrete points or dashed line segments.
2. The `ezplot` command is used for plots of functions. The command `ezplot(f)` plots the function $f(x)$ on the default interval $(-2\pi, 2\pi)$ and the command `ezplot(f,[a,b])` plots the function over the interval $[a,b]$.
3. The commands `plot3` and `ezplot3` are used for three-dimensional plots.
4. The command `meshgrid` is used to generate an xy -grid for surface and contour plots. Specifically the command `[X,Y] = meshgrid(u,v)` transforms the domain specified by vectors u and v into arrays X and Y that can be used for the evaluation of functions of two variables and 3-D surface plots. The rows of the output array X are copies of the vector u and the columns of the output array Y are copies of the vector v . The command can be used with only one input argument in which case `meshgrid(u)` will produce the same arrays as the command `meshgrid(u,u)`.

5. The `mesh` command is used to produce wire frame surface plots and the command `surf` produces a solid surface plot. If $[X, Y] = \text{meshgrid}(u, v)$ and $Z(i, j) = f(u_i, v_j)$, then the command `mesh(X, Y, Z)` will produce a wire frame plot of the function $z = f(x, y)$ over the domain specified by the vectors u and v . Similarly the command `surf(X, Y, Z)` will generate a surface plot over the domain.
6. The MATLAB functions `contour` and `ezcontour` produce contour plots for functions of two variables.
7. The command `meshc` is used to graph both the mesh surface and the contour plot in the same graphics window. Similarly the command `surfc` will produce a surf plot with a contour graph appended.
8. Given an array C whose entries are all real, the command `image(C)` will produce a two-dimensional image representation of the array. Each entry of C will correspond to a small patch of the image. The image array C may be either $m \times n$ or $m \times n \times 3$. If C is an $m \times n$ matrix, then the colors assigned to each patch are determined by MATLAB's current colormap. If C is $m \times n \times 3$, a true color array, then no color map is used. In this case the entries of $C(:, :, 1)$ determine the red intensities of the image, the entries of $C(:, :, 2)$ determine green intensities, and the elements of $C(:, :, 3)$ define the blue intensities.
9. The `colormap` command is used to specify the current colormap for image plots of $m \times n$ arrays.
10. The `imread` command is used to translate a standard graphics file, such as a gif, jpeg, or tiff file, into a true color array. The command can also be used with two output arguments to determine an indexed image representation of the graphics file.

Examples:

1. The graph of the function $f(x) = \cos(x) + \sin^2(x)$ on the interval $(-2\pi, 2\pi)$ can be generated in MATLAB using the following commands:

```
x = -6.3: 0.1: 6.3;
y = cos(x) + sin(x).^2;
plot(x,y)
```

The graph can also be generated using the `ezplot` command. (See Figure 71.2.)

```
f = inline('cos(x) + sin(x).^2')
ezplot(f)
```

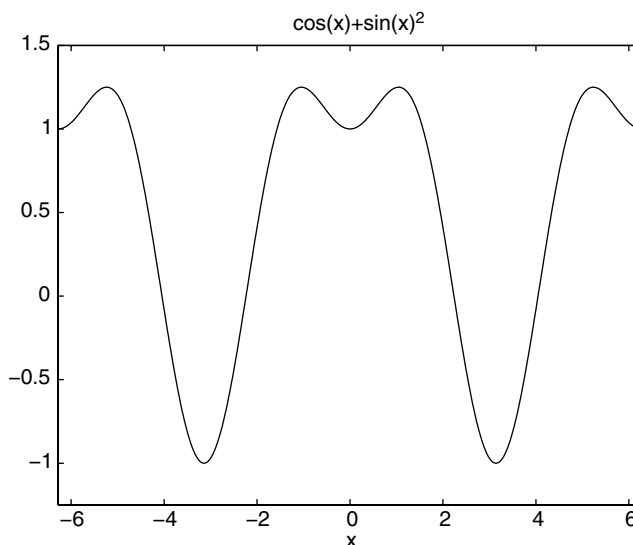


FIGURE 71.2

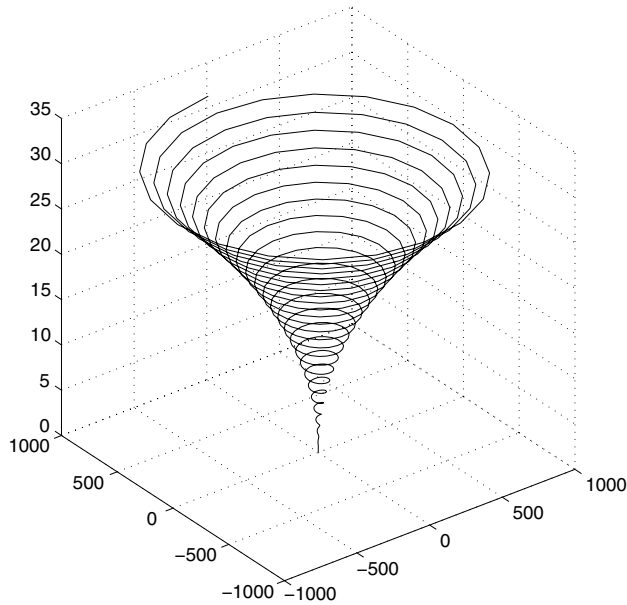


FIGURE 71.3

2. We can generate a three-dimensional plot using the following commands:

```
t = 0:pi/50:10*pi;
plot3(t.^2.*sin(5*t), t.^2.*cos(5*t), t)
grid on
axis square
```

These commands generate the plot shown in Figure 71.3.

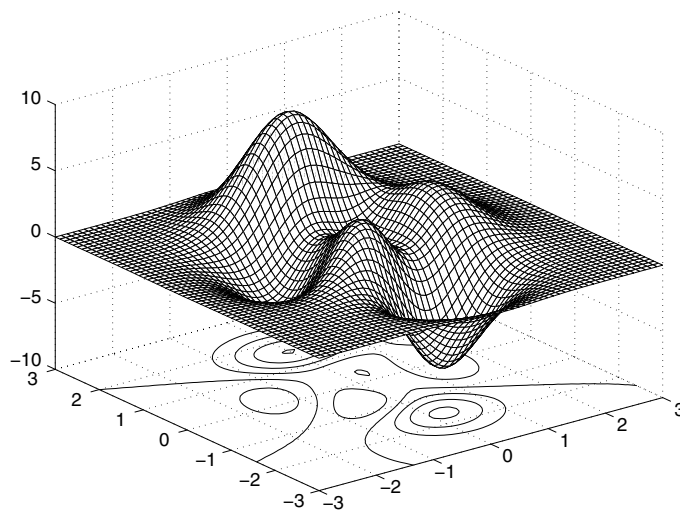


FIGURE 71.4

3. MATLAB's `peaks` function is a function of two variables obtained by translating and scaling Gaussian distributions. The commands

```
[X,Y] = meshgrid(-3:0.1:3);  
Z = peaks(X,Y);  
meshc(X,Y,Z);
```

generate the mesh and contour plots of the `peaks` function. (See Figure 71.4.)

71.10 Symbolic Mathematics in MATLAB

MATLAB's Symbolic Toolbox is based upon the Maple kernel from the software package produced by Waterloo Maple, Inc. The toolbox allows users to do various types of symbolic computations using the MATLAB interface and standard MATLAB commands. All symbolic computations in MATLAB are performed by the Maple kernel. For details of how symbolic linear algebra computations such as matrix inverses and eigenvalues are carried out see Chapter 72.

Facts:

1. MATLAB's symbolic toolbox allows the user to define a new data type, a symbolic object. The user can create symbolic variables and symbolic matrices (arrays containing symbolic variables).
2. The standard matrix operations `+`, `-`, `*`, `^`, `'` all work for symbolic matrices and also for combinations of symbolic and numeric matrices. To add a symbolic matrix and a numeric matrix, MATLAB first transforms the numeric matrix into a symbolic object and then performs the addition using the Maple kernel. The result will be a symbolic matrix. In general if the matrix operation involves at least one symbolic matrix, then the result will be a symbolic matrix.
3. Standard MATLAB commands such as `det`, `inv`, `eig`, `null`, `trace`, `rref`, `rank`, and `sum` work for symbolic matrices.
4. Not all of the MATLAB matrix commands work for symbolic matrices. Commands such as `norm` and `orth` do not work and none of the standard matrix factorizations such as LU or QR work.
5. MATLAB's symbolic toolbox supports variable precision floating arithmetic, which is carried out within the Maple kernel.

Commands:

1. The `sym` command can be used to transform any MATLAB data structure into a symbolic object. If the input argument is a string, the result is a symbolic number or variable. If the input argument is a numeric scalar or matrix, the result is a symbolic representation of the given numeric values.
2. The `syms` command allows the user to create multiple symbolic variables with a single command.
3. The command `subs` is used to substitute for variables in a symbolic expression.
4. The command `colspace` is used to find a basis for the column space of a symbolic matrix.
5. The commands `ezplot`, `ezplot3`, and `ezsurf` are used to plot symbolic functions of one or two variables.
6. The command `vpa(A,d)` evaluates the matrix `A` using variable precision floating point arithmetic with `d` decimal digits of accuracy. The default value of `d` is 32, so if the second input argument is omitted, the matrix will be evaluated with 32 digits of accuracy.

Examples:

1. The command

```
t = sym('t')
```

transforms the string 't' into a symbolic variable t. Once the symbolic variable has been defined, one can then perform symbolic operations. For example, the command

$$\text{factor}(t^2 - 4)$$

will result in the answer

$$(t - 2) * (t + 2).$$

2. The command

$$\text{syms } a \ b \ c$$

creates the symbolic variables a, b, and c. If we then set

$$A = [a, \ b, \ c; \ b, \ c, \ a; \ c, \ a, \ b]$$

the result will be the symbolic matrix

$$A = \begin{bmatrix} a, & b, & c \\ c, & a, & b \\ b, & c, & a \end{bmatrix}.$$

Note that for a symbolic matrix the MATLAB output is in the form of a matrix of row vectors with each row vector enclosed by square brackets.

3. Let A be the matrix defined in the previous example. We can add the 3×3 Hilbert matrix to A using the command $B = A + \text{hilb}(3)$. The result is the symbolic matrix

$$B = \begin{bmatrix} a + 1, & b + 1/2, & c + 1/3 \\ c + 1/4, & a + 1/5, & b + 1/6 \\ b + 1/7, & c + 1/8, & a + 1/9 \end{bmatrix}.$$

To substitute 2 for a in the matrix A we set

$$A = \text{subs}(A, a, 2).$$

The matrix A then becomes

$$A = \begin{bmatrix} 2, & b, & c \\ c, & 2, & b \\ b, & c, & 2 \end{bmatrix}.$$

Multiple substitutions are also possible. To replace b by b + 1 and c by 5, one need only set

$$A = \text{subs}(A, [b, c], [b + 1, 5]).$$

4. If a is declared to be a symbolic variable, the command

$$A = [1 \ 2 \ 1; \ 2 \ 4 \ 2; \ 0 \ 0 \ a]$$

will produce the symbolic matrix

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 6 \\ 0 & 0 & a \end{bmatrix}.$$

The eigenvalues 0, 5, and a are computed using the command `eig(A)`. The command

$$[X, D] = \text{eig}(A)$$

generates a symbolic matrix of eigenvectors

$$X = \begin{bmatrix} -2, & 1/2 * (a + 8)/(-2 + 3 * a), & 1 \\ 1, & & 1, & 2 \\ 0, & 1/2 * a * (a - 5)/(-2 + 3 * a), & 0 \end{bmatrix}$$

and the diagonal matrix

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

When $a = 0$ the matrix A will be defective. One can substitute 0 for a in the matrix of eigenvectors using the command

$$X = \text{subs}(X, a, 0).$$

This produces the numeric matrix

$$X = \begin{bmatrix} -2 & -2 & 1 \\ 1 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

5. If we set $z = \exp(1)$, then MATLAB will compute an approximation to e that is accurate to 16 decimal digits. The command `vpa(z)` will produce a 32 digit representation of z , but only the first 16 digits will be accurate approximations to the digits of e . To compute e more accurately one should apply the `vpa` function to the symbolic expression '`exp(1)`'. The command `z = vpa('exp(1)')` produces an answer $z = 2.7182818284590452353602874713527$, which is accurate to 32 digits.

71.11 Graphical User Interfaces

A graphical user interface (GUI) is a user interface whose components are graphical objects such as pushbuttons, radio buttons, text fields, sliders, checkboxes, and menus. These interfaces allow users to perform sophisticated computations and plots by simply typing numbers into boxes, clicking on buttons, or by moving slidebars.

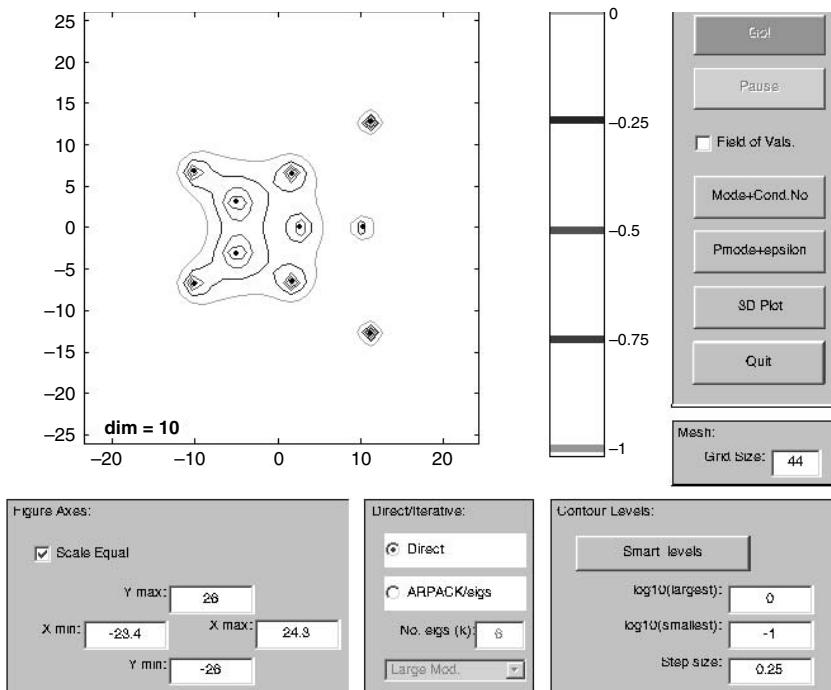


FIGURE 71.5 Eigtool GUI.

Commands:

1. The command guide opens up the MATLAB GUI Design Environment. This environment is essentially a GUI containing tools to facilitate the creation of new GUIs.

Examples:

1. Thomas G. Wright of Oxford University has developed a MATLAB GUI, eigtool, for computing eigenvalues, pseudospectra, and related quantities for nonsymmetric matrices, both dense and sparse. It allows the user to graphically visualize the pseudospectra and field of values of a matrix with just the click of a button.

The *epsilon-pseudospectrum* of a square matrix A is defined by

$$\Lambda_\epsilon(A) = \{z \in \mathbb{C} \mid z \in \sigma(A + E) \text{ for some } E \text{ with } \|E\| \leq \epsilon\}. \quad (71.2)$$

In Figure 71.5 the eigtool GUI is used to plot the epsilon-pseudospectra of a 10×10 matrix for $\epsilon = 10^{-k/4}$, $k = 0, 1, 2, 3, 4$.

For further information, see Chapter 16 and also references [Tre99] and [WT01].

2. The NSF-sponsored ATLAST Project has developed a large collection of MATLAB exercises, projects, and M-files for use in elementary linear algebra classes. (See [LHF03].) The ATLAST M-file collection contains a number of programs that make use of MATLAB's graphical user interface features to present user friendly tools for visualizing linear algebra. One example is the ATLAST cogame utility where students play a game to find linear combinations of two given vectors with the objective of obtaining a third vector that terminates at a given target point in the plane. Students can play the game at any one of four levels or play it competitively by selecting the two person game option. (See Figure 71.6.) At each step of the game a player must enter a pair of coordinates. MATLAB then plots the corresponding linear combination as a directed line segment. The game terminates when

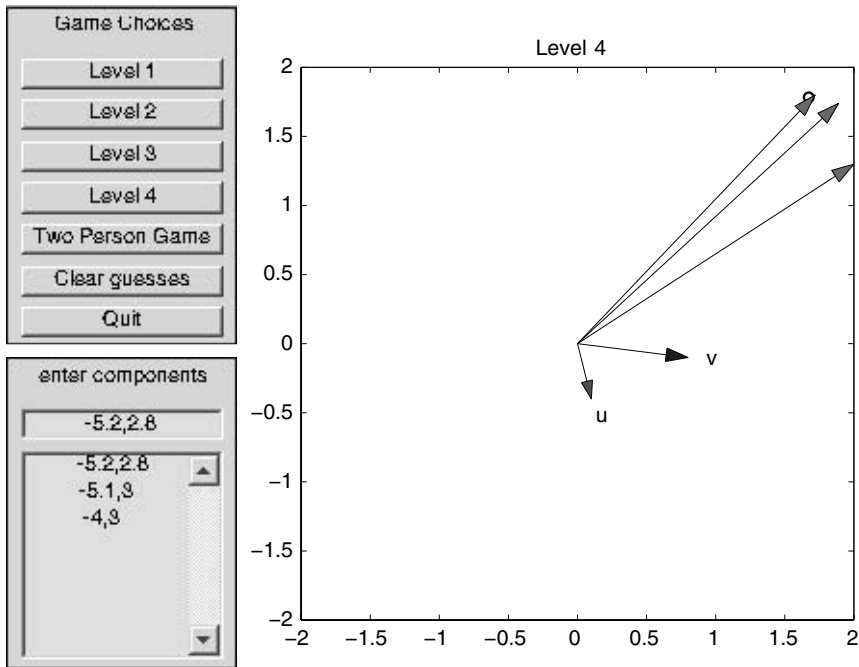


FIGURE 71.6 ATLAST Coordinate Game.

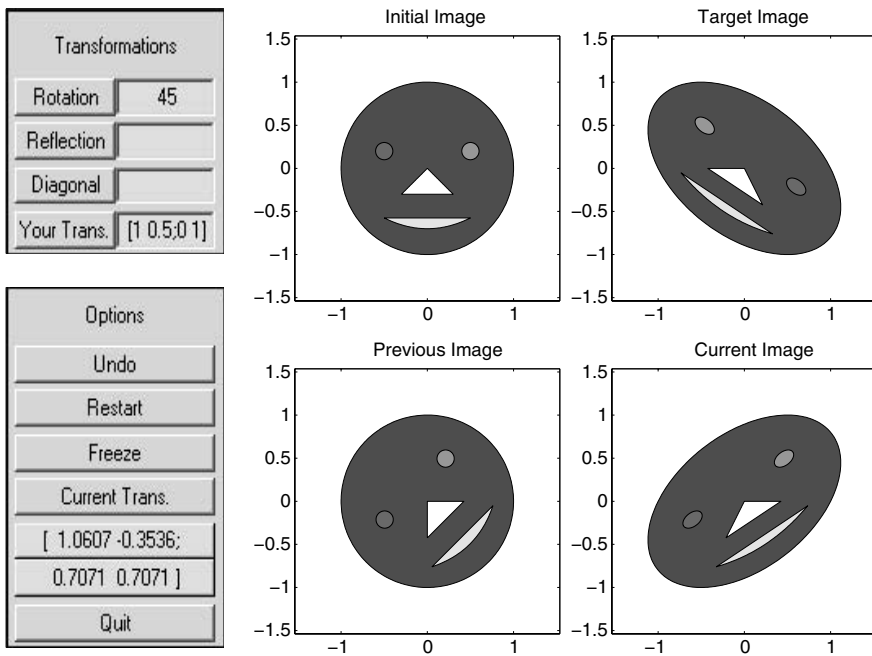


FIGURE 71.7 ATLAST Transformation Utility.

the tip of the plotted line segment lies in the small target circle. A running list of the coordinates entered in the game is displayed in the lower box to the left of the figure. The cogame GUI is useful for teaching lessons on the span of vectors in R^2 and for teaching about different bases for R^2 .

3. The ATLAST transform GUI helps students to visualize the effect of linear transformations on figures in the plane. With this utility students choose an image from a list of figures and then apply various transformations to the image. Each time a transformation is applied, the resulting image is shown in the current image window. The user can then click on the *current transformation* button to see the matrix representation of the transformation that maps the original image into the current image. In Figure 71.7 two transformations were applied to an initial image. First a 45° rotation was applied. Next a transformation matrix $[1, 0; 0.5, 1]$ was entered into the “Your Transformation” text field and the corresponding transformation was applied to the lower left image with the result being displayed in the Current Image window on the lower right. To transform the Current Image into the Target Image directly above it, one would need to apply a reflection transformation.

References

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Linear Algebra in Maple[®]

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72.1 Introduction

Maple[®] is a general purpose computational system, which combines symbolic computation with exact and approximate (floating-point) numerical computation and offers a comprehensive suite of scientific graphics as well. The main library of functions is written in the Maple programming language, a rich language designed to allow easy access to advanced mathematical algorithms. A special feature of Maple is user access to the source code for the library, including the ability to trace Maple's execution and see its internal workings; only the parts of Maple that are compiled, for example, the kernel, cannot be traced. Another feature is that users can link to LAPACK library routines transparently, and thereby benefit from fast and reliable floating-point computation. The development of Maple started in the early 80s, and the company **Maplesoft** was founded in 1988. A strategic partnership with NAG Inc. in 2000 brought highly efficient numerical routines to Maple, including LAPACK.

There are two linear algebra packages in Maple: `LinearAlgebra` and `linalg`. The `linalg` package is older and considered obsolete; it was replaced by `LinearAlgebra` in MAPLE 6. Here we describe only the `LinearAlgebra` package. The reader should be careful when reading other reference books, or the Maple help pages, to check whether reference is made to `vector`, `matrix`, `array` (notice the lower-case initial letter), which means that the older package is being discussed, or to `Vector`, `Matrix`, `Array` (with an upper-case initial letter), which means that the newer package is being discussed.

Facts:

1. Maple commands are typed after a prompt symbol, which by default is “greater than” (`>`). In examples below, keyboard input is simulated by prefixing the actual command typed with the prompt symbol.
2. In the examples below, some of the commands are too long to fit on one line. In such cases, the Maple continuation character backslash (`\`) is used to break the command across a line.
3. Maple commands are terminated by either semicolon (`;`) or colon (`:`). Before Maple 10, a terminator was required, but in the Maple 10 GUI it can be replaced by a carriage return. The semicolon terminator allows the output of a command to be displayed, while the colon suppresses the display (but the command still executes).
4. To access the commands described below, load the `LinearAlgebra` package by typing the command (after the prompt, as shown)


```
> with( LinearAlgebra );
```

If the package is not loaded, then either a typed command will not be recognized, or a different command with the same name will be used.
5. The results of a command can be assigned to one or more variables. Thus,


```
> a := 1 ;
```

assigns the value 1 to the variable a , while

```
> (a,b,c) := 1,2,3 ;
```

assigns a the value 1, b the value 2 and c the value 3. *Caution:* The operator colon-equals (`:=`) is assignment, while the operator equals (`=`) defines an equation with a left-hand side and a right-hand side.
6. A sequence of expressions separated by commas is an expression sequence in Maple, and some commands return expression sequences, which can be assigned as above.
7. Ranges in Maple are generally defined using a pair of periods (`..`). The rules for the ranges of subscripts are given below.

72.2 Vectors

Facts:

1. In Maple, vectors are not just lists of elements. Maple separates the idea of the mathematical object `Vector` from the data object `Array` (see Section 72.4).
2. A Maple `Vector` can be converted to an `Array`, and an `Array` of appropriate shape can be converted to a `Vector`, but they cannot be used interchangeably in commands. See the help file for `convert` to find out about other conversions.
3. Maple distinguishes between column vectors, the default, and row vectors. The two types of vectors behave differently, and are not merely presentational alternatives.

Commands:

1. Generation of vectors:
 - `Vector( [x1, x2, ...] )` Construct a column vector by listing its elements. The length of the list specifies the dimension.
 - `Vector[column]( [x1, x2, ... ] )` Explicitly declare the column attribute.
 - `Vector[row]( [x1, x2, ... ] )` Construct a row vector by initializing its elements from a list.
 - `<v1, v2, ...>` Construct a column vector with elements v_1, v_2 , etc. An element can be another column vector.

- $\langle v_1 | v_2 | \dots \rangle$ Construct a row vector with elements v_1, v_2 , etc. An element can be another row vector. A useful mnemonic is that the vertical bars remind us of the column dividers in a table.
- `Vector( n, k -> f(k) )`. Construct an n -dimensional vector using a function $f(k)$ to define the elements. $f(k)$ is evaluated sequentially for k from 1 to n . The notation $k \rightarrow f(k)$ is Maple syntax for a univariate function.
- `Vector( n, fill=v )` An n -dimensional vector with every element v .
- `Vector( n, symbol=v )` An n -dimensional vector containing symbolic components v_k .
- `map( x -> f(x), V )` Construct a new vector by applying function $f(x)$ to each element of the vector named V . *Caution:* the command is `map` not `Map`.

2. Operations and functions:

- `v[i]` Element i of vector $\mathbf{v}$. The result is a scalar. *Caution:* A symbolic reference `v[i]` is typeset as v_i on output in a Maple worksheet.
- `v[p..q]` Vector consisting of elements v_i , $p \leq i \leq q$. The result is a `Vector`, even for the case `v[p..p]`. Either of p or q can be negative, meaning that the location is found by counting backwards from the end of the vector, with -1 being the last element.
- `u+v`, `u-v` Add or subtract `Vectors` $\mathbf{u}$, $\mathbf{v}$.
- `a*v` Multiply vector $\mathbf{v}$ by scalar a . Notice the operator is “asterisk” (*).
- `u . v`, `DotProduct( u, v )` The inner product of `Vectors` $\mathbf{u}$ and $\mathbf{v}$. See examples for complex conjugation rules. Notice the operator is “period” (.) not “asterisk” (*) because inner product is not commutative over the field of complex numbers.
- `Transpose( v )`, `v^%T` Change a column vector into a row vector, or *vice versa*. Complex elements are not conjugated.
- `HermitianTranspose( v )`, `v^%H` Transpose with complex conjugation.
- `OuterProductMatrix( u, v )` The outer product of `Vectors` $\mathbf{u}$ and $\mathbf{v}$ (ignoring the row/column attribute).
- `CrossProduct( u, v )`, `u &x v` The vector product, or cross product, of three-dimensional vectors $\mathbf{u}$, $\mathbf{v}$.
- `Norm( v, 2 )` The 2-norm or Euclidean norm of vector $\mathbf{v}$. Notice that the second argument, namely the 2, is necessary, because `Norm( v )` defaults to the infinity norm, which is different from the default in many textbooks and software packages.
- `Norm( v, p )` The p -norm of $\mathbf{v}$, namely $(\sum_{i=1}^n |v_i|^p)^{1/p}$.

Examples:

In this section, the imaginary unit is the Maple default I . That is, $\sqrt{-1} = I$. In the matrix section, we show how this can be changed. To save space, we shall mostly use row vectors in the examples.

1. Generate vectors. The same vector created different ways.

```
> Vector[row]([0,3,8]): <0|3|8>: Transpose(<0,3,8>): Vector[row]
(3,i->i^2-1);
```

$$[0,3,8]$$

2. Selecting elements.

```
> V:=<a|b|c|d|e|f>: V1:=V[2..4]; V2:=V[-4..-1];
V3:=V[-4..4];
```

$$V1 := [b, c, d], \quad V2 := [c, d, e, f], \quad V3 := [c, d]$$

3. A Gram–Schmidt exercise.

```
> u1 := <3|0|4>: u2 := <2|1|1>: w1n := u1/Norm( u1, 2 );
```

$$w1n := [3/5, 0, 4/5]$$

```
> w2 := u2 - (u2 . w1n)*w1n; w2n := w2/Norm( w2, 2 );
```

$$w2n := \left[\frac{2\sqrt{2}}{5}, \frac{\sqrt{2}}{2}, -\frac{3\sqrt{2}}{10} \right]$$

4. Vectors with complex elements. Define column vectors $\mathbf{u}_c, \mathbf{v}_c$ and row vectors $\mathbf{u}_r, \mathbf{v}_r$.

```
> uc := <1 + I, 2>: ur := Transpose( uc ): vc := <5, 2 - 3*I>:
vr := Transpose( vc ):
```

The inner product of column vectors conjugates the first vector in the product, and the inner product of row vectors conjugates the second.

```
> inner1 := uc . vc; inner2 := ur . vr;
      inner1 := 9-11 I , inner2 := 9+11 I
```

Maple computes the product of two similar vectors, i.e., both rows or both columns, as a true mathematical inner product, since that is the only definition possible; in contrast, if the user mixes row and column vectors, then Maple does not conjugate:

```
> but := ur . vc;
```

$$but := 9 - I$$

Caution: The use of a period (.) with complex row and column vectors together differs from the use of a period (.) with complex $1 \times m$ and $m \times 1$ matrices. In case of doubt, use matrices and conjugate explicitly where desired.

72.3 Matrices

Facts:

1. One-column matrices and vectors are not interchangeable in Maple.
2. Matrices and two-dimensional arrays are not interchangeable in Maple.

Commands:

1. Generation of Matrices.

- `Matrix( [[a,b,...], [c,d,...], ... ] )` Construct a matrix row-by-row, using a list of lists.
- `<< a|b|...>, <c|d|...>, ...>` Construct a matrix row-by-row using vectors. Notice that the rows are specified by row vectors, requiring the `|` notation.
- `<< a,b,...>|<c,d,...>|...>` Construct a matrix column-by-column using vectors. Notice that each vector is a column, and the columns are joined using `|`, the column operator.

Caution: Both variants of the `<< ... >>` constructor are meant for interactive use, not programmatic use. They are slow, especially for large matrices.

- `Matrix( n, m, (i, j) -> f(i, j) )` Construct a matrix $n \times m$ using a function $f(i, j)$ to define the elements. $f(i, j)$ is evaluated sequentially for i from 1 to n and j from 1 to m . The notation $(i, j) -> f(i, j)$ is Maple syntax for a bivariate function $f(i, j)$.
- `Matrix( n, m, fill=a )` An $n \times m$ matrix with each element equal to a .
- `Matrix( n, m, symbol=a )` An $n \times m$ matrix containing subscripted entries a_{ij} .
- `map( x -> f(x), M )` A matrix obtained by applying $f(x)$ to each element of M .
[Caution: the command is `map` not `Map`.]
- `<< A|B>, < C|D>>` Construct a partitioned or block matrix from matrices A, B, C, D . Note that `< A|B >` will be formed by adjoining columns; the block `< C|D >` will be placed below `< A|B >`. The Maple syntax is similar to a common textbook notation for partitioned matrices.

2. Operations and functions

- `M[i, j]` Element i, j of matrix M . The result is a scalar.
- `M[1..-1, k]` Column k of Matrix M . The result is a Vector.
- `M[k, 1..-1]` Row k of Matrix M . The result is a row Vector.
- `M[p..q, r..s]` Matrix consisting of submatrix m_{ij} , $p \leq i \leq q$, $r \leq j \leq s$. In HANDBOOK notation, $M[\{p, \dots, q\}, \{r, \dots, s\}]$.
- `Transpose( M )`, `M^T` Transpose matrix M , without taking the complex conjugate of the elements.
- `HermitianTranspose( M )`, `M^H` Transpose matrix M , taking the complex conjugate of elements.
- `A ± B` Add/subtract compatible matrices or vectors A, B .
- `A.B` Product of compatible matrices or vectors A, B . The examples below detail the ways in which Maple interprets products, since there are differences between Maple and other software packages.
- `MatrixInverse( A )`, `A^(-1)` Inverse of matrix A .
- `Determinant( A )` Determinant of matrix A .
- `Norm( A, 2 )` The (subordinate) 2-norm of matrix A , namely $\max_{\|u\|_2=1} \|Au\|_2$ where the norm in the definition is the vector 2-norm.

Cautions:

- (a) Notice that the second argument, i.e., 2, is necessary because `Norm( A )` defaults to the infinity norm, which is different from the default in many textbooks and software packages.
 - (b) Notice also that this is the largest *singular value* of A , and is usually different from the Frobenius norm $\|A\|_F$, accessed by `Norm( A, Frobenius )`, which is the Euclidean norm of the vector of elements of the matrix A .
 - (c) Unless A has floating-point entries, this norm will not usually be computable explicitly, and it may be expensive even to try.
- `Norm( A, p )` The (subordinate) matrix p -norm of A , for integers $p \geq 1$ or for p being the symbol `infinity`, which is the default value.